

Piper Seminole

Training Supplement

ATPFlightSchool.com

Revised 2025-11-06

IMPORTANT NOTICE

Refer to POH/AFM

Do not use procedures listed without referencing the full procedures described in the approved Owner's Manual, POH, or POH/AFM specific to the airplane you are flying. Endurance and fuel capacities may vary considerably depending on the specific model / serial number being flown and any modifications it may have.

Copyright © 2009–2025 ATP USA, Inc.

The content of this manual is furnished for informational use only, and is subject to change without notice. Airline Transport Professionals assumes no responsibility or liability for any errors or inaccuracies that may appear in this manual. This manual does not replace the Piper Seminole Pilot Operating Handbook, FAA Airplane Flying Handbook, or Practical Test Standards / Airman Certification Standards. Nothing in this manual shall be interpreted as a substitute for the exercise of sound judgement.

No part of this publication may be reproduced, stored in a retrieval system, or transmitted, in any form or by any means electronic, mechanical or otherwise, without the prior written permission of Airline Transport Professionals.

To view recent changes to this supplement, visit:
atpflightschool.com/changes/supp-seminole

Introduction

Thank you for choosing ATP for your flight program.

ATP flies over 540,000 hours per year to provide over 12,500 FAA certificates annually. ATP is committed to providing you low prices; efficient, standardized, and effective flight programs; and–most importantly–the highest level of safety.

To maximize the effectiveness of your flight program, ATP's *Piper Seminole Training Supplement* contains a condensed overview of multi-engine aerodynamics, portions of the Piper Seminole POH, and flight procedures. You must have a complete knowledge of all information contained in this supplement prior to the start of your program. This information will assist you with your training and flight check.

It is critical that you **memorize** the following:

- Emergency Engine Failure Checklists.
- V-Speeds.
- Answers to all questions contained in the "Oral Review" section in the back of the supplement.

The information in this supplement is highly condensed and serves as a good quick reference, but it must not be used as a substitute for the FAA-approved pilot's operating handbook required for safe operation of the airplane.

Thank you for giving ATP an opportunity to serve you.

Contents

Engine-Out Aerodynamics 1
- Aerodynamic Effects of an Engine Failure 1
- Engine Inoperative Climb Performance 2
- Approximate Drag Factors per the Piper Seminole POH 2
- Airspeeds for Max Single-Engine Performance 3
- Sideslip Versus Zero Sideslip 3
- Single-Engine Service Ceiling 4
- Single-Engine Absolute Ceiling 4
- Climb Performance Depends on Four Factors 4
- Critical Engine 5
- V_{MC} 7

Aircraft Systems 9
- Engines 9
- Propellers 10
- Landing Gear 12
- Brakes 13
- Flaps 13
- Vacuum Pumps 14
- Pitot Static 14
- Fuel System 14
- Electrical System 15
- Heater 15
- Stall Warning Horn 16
- Emergency Exit 16
- Inoperative Instruments & Equipment per FAR 91.213 17
- Lynx NGT-9000 17
- Garmin G500 18
- G500 Standby Instruments 20

Aircraft Systems (Continued)
- G500 Failures & Partial-Panel Approaches 21
- Garmin G1000 24
- Garmin G5 31
- Standard Avionics Configurations 32
- Dual Garmin GNS430 (with or without Lynx NGT-9000) 38
- GPS Setup for VFR Airwork & Maneuvers 39
- List of Pilots Guides 39
- Notable Differences Between Seminoles by Model Year 40

Performance / Weight & Balance 41
- Piper Seminole V-Speeds 41
- Performance Charts 42
- Sample Weight & Balance Problem 42
- Formulas 43
- CG Envelope Graph 44

Departure Procedures 45
- Passenger Briefing 45
- Pre-Takeoff Briefing (Standard Procedures) 45
- Normal Takeoff (Flaps 0°) 46
- Short Field Takeoff & Climb (Flaps 0°) 47
- Engine Failure Procedure 48

Arrival Procedures 50
- Approach Briefing 50
- Announced Callouts on Approach 50
- Summary of Callouts 52
- Stabilized Approaches 53
- Go Around Philosophy 53
- Collision Avoidance 54

Arrival Procedures (Continued)
Normal Landing Technique 55
Normal Visual Approach & Landing 56
Short Field Visual Approach & Landing .. 58
Precision Approach One or Two Engines 59
Non-Precision Straight-In Approach One or Two Engines 60
Circle to Land Approaches Two Engines 61
Go Around / Missed Approach 62
Holding 63
Approach, Procedure Turn & Holding Power Settings 63

In-Flight Maneuvers & Procedures 64
Spin Awareness 64
In-Flight Maneuvers 65
Clearing Turns for In-Flight Maneuvers .. 65
Standard Avionics Configurations 65
V_{MC} Demonstration 66
Maneuvering with One Engine Inoperative 67
In-Flight Engine Failure Demo 67

In-Flight Maneuvers & Procedures (Continued)
Airstart Procedure 70
Recovery from Unusual Attitudes 71
Steep Turns 71
Maneuvering During Slow Flight 72
Stall Recovery 73
Power-Off Stall 74
Power-On Stall 75
Steep Turns 76
Maneuvering During Slow Flight 77
Power-Off Stall 77
Power-On Stall 78
Accelerated Stall 79
Emergency Descent 79
Airspeed Drag Demonstration 80
Steep Turns 81
Approach to Stall in Approach Configuration (Clean) 81
Approach to Stall in Landing Configuration 82

Oral Review 83
Sample Oral Questions 84

SECTION 1

ENGINE-OUT AERODYNAMICS

Aerodynamic Effects of an Engine Failure

When an engine failure occurs in a multi-engine aircraft, asymmetric thrust and drag cause the following effects on the aircraft's axes of rotation:

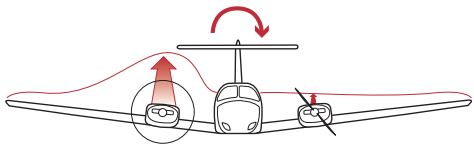
Pitch Down (Lateral Axis)

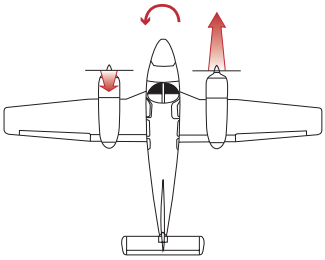
Loss of accelerated slipstream over the horizontal stabilizer causes it to produce less negative lift, causing the aircraft to pitch down. To compensate for the pitch down effect, additional back pressure is required.

> ⓘ The Seminole has a lesser pitch-down effect than most light twins because the T-tail configuration removes the horizontal stabilator from the accelerated slipstream.

Roll Toward the Failed Engine (Longitudinal Axis)

The wing produces less lift on the side of the failed engine due to the loss of accelerated slipstream. Reduced lift causes a roll toward the failed engine and requires additional aileron deflection into the operating engine.





Yaw Toward the Dead Engine (Vertical Axis)

Loss of thrust and increased drag from the windmilling propeller cause the aircraft to yaw toward the failed engine. This requires additional rudder pressure on the side of the operating engine. "Dead foot, dead engine."

Engine Inoperative Climb Performance

Climb performance depends on the excess power needed to overcome drag. When a multi-engine airplane loses an engine, the airplane loses 50% of its available power. This power loss results in a loss of approximately 80% of the aircraft's excess power and climb performance. Drag is a major factor relative to the amount of excess power available. An increase in drag (such as the loss of one engine) must be offset by additional power. This additional power is now taken from the excess power, making it unavailable to aid the aircraft in climb. When an engine is lost, maximize thrust (full power) and minimize drag (flaps and gear up, prop feathered, etc.) in order to achieve optimum single-engine climb performance.

Approximate Drag Factors per the Piper Seminole POH

1. Flaps 25°.....**-240 FPM**
2. Flaps 40°.....**-275 FPM**
3. Windmilling Prop .....**-200 FPM**
4. Gear Extended .....**-250 FPM**

ATP's experience indicates more than -200 FPM drag from the windmilling prop.

Under FAR Part 23:

The FAA does not require multi-engine airplanes that weigh less than 6,000 pounds or have a V_{SO} speed under 61 knots to meet any specified single-engine performance criteria. No single engine climb performance is required. Actual climb performance is documented by the manufacturer.

The Seminole's max TOW is 3,800 pounds and V_{SO} speed is 55 KIAS. Review the POH for single-engine climb performance for specific conditions.

Airspeeds for Max Single-Engine Performance

V_{XSE}

The airspeed for the steepest angle of climb on single-engine.

V_{YSE}

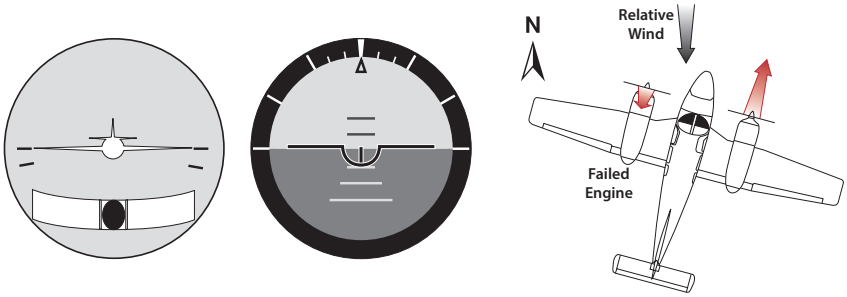
The airspeed for the best rate of climb on single-engine. (Or for the slowest loss of altitude on drift-down.) Blueline is the marking on the airspeed indicator corresponding to V_{YSE} at max weight.

Sideslip Versus Zero Sideslip

During flight with one engine inoperative, proper pilot technique is required to maximize aircraft performance. An important technique is to establish a Zero Sideslip Condition.

Sideslip Condition (Undesirable)

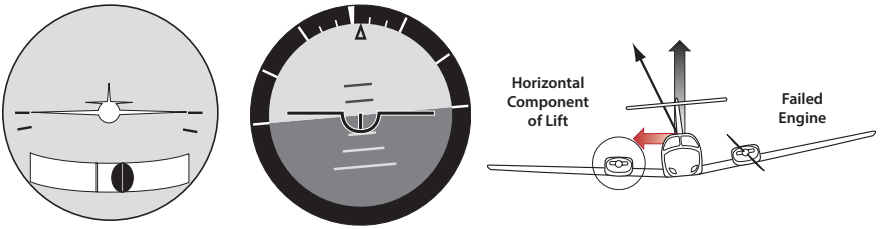
When an engine failure occurs, thrust from the operating engine yaws the aircraft. To maintain aircraft heading with the wings level, rudder must be applied toward the operating engine. This rudder force results in the sideslip condition by moving the nose of the aircraft in a direction resulting in the misalignment of the fuselage and the relative wind. This condition usually allows the pilot to maintain aircraft heading; however, it produces a high drag condition that significantly reduces aircraft performance.



Sideslip Condition:
Ball Centered & Wings Level, Aircraft Tracking North

Zero Sideslip Condition (Best Performance)

The solution to maintaining aircraft heading and reducing drag to improve performance is the Zero Sideslip Condition. When the aircraft is banked into the operating engine (usually 2°-5°), the bank angle creates a horizontal component of lift. The horizontal lift component aids in counteracting the turning moment of the operating engine, minimizing the rudder deflection required to align the longitudinal axis of the aircraft to the relative wind. In addition to banking into the operating engine, the appropriate amount of rudder required is indicated by the inclinometer ball being "split" towards the operating engine side. The Zero Sideslip Condition aligns the fuselage with the relative wind to minimize drag and must be flown for optimum aircraft performance.



Zero Sideslip Condition:
2°-5° Bank into Operating Engine

Single-Engine Service Ceiling

Single-engine service ceiling is the maximum density altitude at which the single-engine best rate of climb airspeed (V_{YSE}) will produce a 50 FPM rate of climb with the critical engine inoperative.

Single-Engine Absolute Ceiling

Single-engine absolute ceiling is the maximum density altitude that an aircraft can attain or maintain with the critical engine inoperative. V_{YSE} and V_{XSE} are equal at this altitude. The aircraft drifts down to this altitude when an engine fails.

Climb Performance Depends on Four Factors

- ***Airspeed:*** Too little or too much will decrease climb performance.
- ***Drag:*** Gear, Flaps, Cowl Flaps, Flight Control Deflection, Prop, and Sideslip.
- ***Power:*** Amount available in excess of that needed for level flight. *(Engines may require leaning due to altitude for max engine performance.)*
- ***Weight:*** Passengers, baggage, and fuel load greatly affect climb performance.

Critical Engine

The critical engine is the engine that, when it fails, most adversely affects the performance and handling qualities of the airplane.



The Seminole is equipped with a counter-rotating propeller on the right engine. The failure of either engine has the same effects on performance. This is why the Seminole does not have a critical engine.

On most multi-engine aircraft, both propellers rotate clockwise as viewed from the cockpit. By understanding the following factors when flying an aircraft that has both propellers rotating clockwise, it will be apparent that a left-engine failure makes the aircraft more difficult to fly than a right-engine failure. The clockwise rotation of the props contributes to the following factors that cause the left engine to be critical:

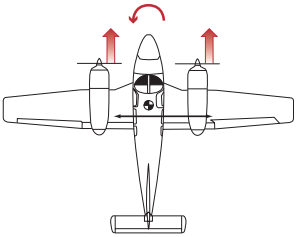
P P-Factor

A Accelerated Slipstream

S Spiraling Slipstream

T Torque

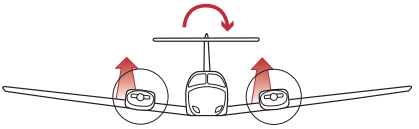
P-Factor (Yaw)



Both propellers turn clockwise as viewed from the cockpit. At low airspeeds and high angles of attack, the descending blade produces more thrust than the ascending blade due to its increased angle of attack. Though both propellers produce the same overall thrust, the descending blade on the right engine has a longer arm from the CG (or greater leverage) than the descending blade on the left engine. The left engine produces the thrust closest to center line. The yaw produced by the loss of the left engine will be greater than the yaw produced by the loss of the right engine, making the left engine critical.

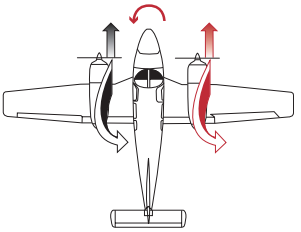
Accelerated Slipstream (Roll and Pitch)

P-Factor causes more thrust to be produced on the right side of the propeller. This yields a center of lift that is closer to the aircraft's longitudinal axis on the left engine and further from the longitudinal axis on the right engine and also results in less negative lift on the tail. Because of this, the roll produced by the loss of the left engine will be greater than the roll produced by the loss of the right engine, making the left engine critical.



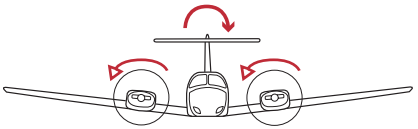
Spiraling Slipstream (Yaw)

A spiraling slipstream from the left engine hits the vertical stabilizer from the left, helping to counteract the yaw produced by the loss of the right engine. However, with a left engine failure, slipstream from the right engine does not counteract the yaw toward the dead engine because it spirals away from the tail, making the left engine critical.



Torque (Roll)

For every action, there is an equal and opposite reaction. Since the propellers rotate clockwise, the aircraft will tend to roll counterclockwise. When the right engine is lost, the aircraft will roll to the right. The right rolling tendency, however, is reduced by the torque created by the left engine. When the left engine is lost, the aircraft will roll to the left, and the torque produced by the right engine will add to the left rolling tendency requiring more aileron input, which increases drag, making the left engine critical.



Summary

On most light multi-engine aircraft when the critical engine is inoperative, both directional control and performance suffer more than when the non-critical engine is inoperative.

V_{MC}

V_{MC} is the calibrated airspeed at which, following the sudden critical loss of thrust, it is possible to maintain control of the airplane. For multi-engine airplanes, the pilot must determine V_{MC}, for the most critical configurations used in takeoff and landing operations.

When the PA-44 was originally designed, the historic regulation 14 CFR Part 23.149 outlined the items that would most accurately simulate critical engine failure and represent the most critical mode of powerplant failure expected with respect to controllability. You can use the acronym SMACFUM to explain this.

- Standard day conditions at sea level (max engine power)
- Most unfavorable weight
- Aft CG (or most unfavorable)
- Critical engine windmilling
- Flaps in the takeoff position & Landing gear retracted
- Up to 5° bank into the operative engine
- Max available takeoff power on each engine

Any change to the above condition will change V_{MC}, possibly significantly. The following summarizes how V_{MC} may be affected by the above conditions.

1. Standard Day Sea Level

Standard conditions yield high air density that allows the engine to develop maximum power. An increase in altitude or temperature (a decrease in air density) will result in reduced engine performance and prop efficiency. This decreases the adverse yaw effect. V_{MC} speed decreases as altitude increases.

2. Most Unfavorable Weight and Center of Gravity

The certification test allows up to 5° bank into the operating engine. In a given bank, the heavier the aircraft, the greater the horizontal component of lift that adds to the rudder force. As weight increases, the horizontal component of lift increases, which added to the rudder, decreases V_{MC}. As the center of gravity moves forward, the moment arm between the rudder and the CG is lengthened, increasing the leverage of the rudder. This increased leverage increases the rudder's effectiveness and results in a lower V_{MC} speed.

3. Critical Engine Prop Windmilling

When the propeller is in a low pitch position (unfeathered), it presents a large area of resistance to the relative wind. This resistance causes the engine to "windmill." The windmilling creates a large amount of drag and results in a yawing moment into the dead engine. When the propeller is "feathered," the blades are in a high pitch position, which aligns them with the relative wind, minimizing drag. A feathered prop will decrease drag and lower V_{MC}.

4. Flaps Takeoff Position, Landing Gear Up, Trimmed for Takeoff

When the gear is extended, the gear and gear doors have a keel effect, reducing the yawing tendency and decreasing the V_{MC} speed. Extended flaps have a stabilizing effect that may reduce V_{MC} speed.

5. Up to 5° Bank into the Operating Engine

When the wings are level, only the rudder is used to stop the yaw produced by the operating engine (sideslip condition). Banking into the operating engine creates a horizontal component of lift which aids the rudder force. With this horizontal component of lift and full rudder deflection, V_{MC} is at the lowest speed. V_{MC} increases with decreasing bank by a factor of approximately 3 knots per degree of bank angle.

6. Maximum Power On The Operating Engine

When the operating engine develops maximum power, adverse yaw is increased toward the inoperative engine. The pilot must overcome this yaw to maintain directional control. Any condition that increases power on the operating engine will increase V_{MC} speed. Any condition that decreases power on the operating engine (such as power reduction by the pilot, an increase in altitude, temperature, low density, or aging engine) will decrease V_{MC}.



At V_{MC}, the rudder pedal force required to maintain control must not exceed 150 pounds and it must not be necessary to reduce power of the operative engine(s). During the maneuver, the airplane must not assume any dangerous attitude and it must be possible to prevent a heading change of more than 20°.



Piper determined V_{MC} for the Seminole at 1,500' MSL with a weight of 2,730 lbs. and 5° bank into the operating engine. The test results were then computed to sea level conditions.

NOTE: Each aircraft is different and may be subject to different handling qualities than discussed here. Recovery from loss of directional control should always follow the guidelines of the POH and FAA Airplane Flying Handbook.

Engines

The Seminole is equipped with two Lycoming, 4-cylinder, O-360 (opposed, 360 cubic inch) engines rated at 180 horsepower at 2700 RPM. The right engine is designated as an LO-360 due to the fact that it rotates to the left. The engines are direct drive (crankshaft connected directly to the propeller), horizontally opposed (pistons oppose each other), piston driven and normally aspirated (no turbo or supercharging). Seminoles with model years prior to 2022 have carbureted engines, and those with model years from 2022 onwards have fuel-injected engines.

- **L** Lycoming
- **H** Horizontally Opposed
- **A** Air Cooled
- **N** Normally Aspirated
- **D** Direct Drive

NOTE: Certain procedures, particularly those for engine starting, are substantially different between carbureted and fuel-injected engines. Be sure to use the correct checklist for each specific aircraft.

Ignition

Engine ignition is provided through the use of two magnetos per engine. As originally equipped, both are conventional engine-driven magnetos which are independent of the aircraft's electrical system and each other.

In some aircraft, one magneto per engine has been replaced with an EIS (Electronic Ignition System) solid-state magneto. This magneto is powered by the main battery. EIS magnetos are more reliable, while the remaining conventional

magneto provides redundancy in the event of electrical system failure. Aircraft with EIS magnetos can be identified by a placard on the instrument panel advising not to operate the aircraft with low battery voltage. There are no operational differences between aircraft with an EIS magneto and those with only conventional magnetos.

Carburetor Icing

For aircraft with carbureted engines, under certain moist atmospheric conditions at temperatures of 20° to 70° F (-5° to 20° C), it is possible for ice to form in the induction system, even in summer weather. This is due to the high air velocity through the carburetor venturi and the absorption of heat from this air by vaporization of the fuel. To avoid this, the carburetor heat is provided to replace the heat lost by vaporization. The initial signs of carburetor ice can include engine roughness and a drop in manifold pressure. Carburetor heat should be selected on if carburetor ice is encountered. Adjust mixture for maximum smoothness.

NOTE: Carburetor heat also serves as an alternate air source in case the engine's air intake is blocked. In lieu of carburetor heat, fuel-injected Seminoles have a similar "Alternate Air" control that provides heated, unfiltered air to the induction system.

Propellers

The Seminole is equipped with Hartzell two-bladed, controllable pitch, constant speed, full feathering metal propellers.

Controllable Pitch

Controllable pitch is the ability to control engine RPM by varying the pitch of the propeller blades. When the blue propeller control handle is moved forward, oil pressure, regulated by a propeller governor, drives a piston, which moves the blades to a low pitch–high RPM (unfeathered) position. When the blue propeller control handle is moved aft, oil pressure is reduced by the propeller governor. This allows a nitrogen-charged cylinder, spring, and centrifugal counterweights to drive the blades to a high pitch–low RPM (feathered) position.

Constant Speed

After RPM setting is selected with the blue propeller control handles, the propeller governor will automatically vary oil pressure inside the propeller hub to change the propeller blade pitch in order to maintain a constant engine RPM. Because of this, changes in power setting (manifold pressure) and flight attitude will not cause a change in RPM.

Full Feathering

When the propeller blades are in alignment with the relative wind, they are feathered. Feathered propeller blades reduce the drag caused by the blade area exposed to the relative wind. Feathering the propeller blades on the Seminole is accomplished by moving the blue propeller control handle fully aft past the low RPM detent, into the “FEATHER” position. The propeller takes approximately six seconds to feather. When feathering the propeller, the mixture should be placed to cutoff to stop engine combustion and power production.

The Seminole is equipped with a centrifugal stop pin that prevents propeller feathering below 950 RPM. The purpose of this is to allow the propeller blades to remain in a low pitch upon engine shutdown. This will prevent excessive loads on the engine starter during the next engine start.

Regardless of the Prop Lever position, if oil pressure is lost, the propeller will feather when the RPM is above 950 RPM. Typically, RPM will be above 950 in flight and on takeoff roll and landing roll due to airflow over the propeller.

Propeller Overspeed

Propeller overspeed is usually caused by a malfunction in the propeller governor which allows the propeller blades to rotate to full low pitch. If propeller overspeed should occur, retard the throttle. The propeller control should be moved to full “DECREASE RPM” and then set if any control is available. Airspeed should be reduced and throttle used to maintain a maximum of 2700 RPM.

Propeller and Governor Ground Checks

As part of the Run Up checklist, the propeller and propeller governor systems are dynamically tested. This consists of two steps: *Prop Feather* and *Prop Exercise* and *Governor Function*.

The *Prop Feather* check confirms that the propeller is able to move toward the feathered position, which is critical in an engine failure situation where the drag of an unfeathered propeller would be detrimental. When the propeller control lever is pulled back, the propeller governor decreases oil pressure supplied to the propeller hub. This allows the propeller's internal spring, counterweights, and nitrogen pressure to increase the propeller blade angle, which decreases engine RPM. Additionally, it allows the relatively cold and thick oil in the propeller's hub to be exchanged with the engine's warmer oil ensuring the propeller responds in a normal manner.

The *Prop Exercise and Governor Function* check also circulates warm oil into the propeller's hub and provides confirmation that the propeller governor is able to maintain engine RPM. As RPM is reduced by the governor's commanded increase

in propeller blade angle, an associated slight increase in manifold pressure can be observed.

Landing Gear

The Seminole is equipped with hydraulically actuated, fully retractable, tricycle-type landing gear. Hydraulic pressure for gear operation is provided by an electrically powered, reversible hydraulic pump. The gear is held in the up position solely by hydraulic pressure. Springs assist in gear extension and in locking the gear in the down position. After the gear is down and the downlock hooks engage, springs maintain force on each hook to keep it locked until it is released by applying hydraulic pressure with the gear selector.

In conventional and G500 Seminoles, the position of the gear is indicated by a combination of:

- Gear indicator lights. These three green lights (one for each gear) are directly above the landing gear selector switch. They illuminate when the down limit switches are depressed, indicating that the gear is down and locked.
- The "GEAR UNSAFE" annunciator. This red annunciator at the top of the instrument panel illuminates when neither the up nor down limit switches are depressed, usually indicating that the gear is in transit.

In G1000 Seminoles, gear status is displayed on the MFD:

- Green circles: gear down and locked
- Hollow white circles: gear up
- Crosshatched squares: gear in transit
- Red circles: abnormal / unknown gear position

Additionally, a convex mirror on the left engine nacelle allows the pilot to visually confirm the position of the nose gear.

A gear warning system is activated under any of the following conditions:

1. The gear is not locked down with the throttle lever positioned below approximately 15" manifold pressure (MP) on one or both engines.
2. The gear is not locked down with wing flaps selected to 25° or 40°.
3. The gear handle is in the up position on the ground (tested only by authorized maintenance personnel).

This warning consists of a rapid (90 or 180 cycles per minute) beeping in conventional or G500 Seminoles, and an aural *"CHECK GEAR"* voice alert (accompanied by a CAS warning) in G1000 Seminoles.

Gear retraction on the ground is prevented by a squat switch located on the left main landing gear. On the ground, the switch is open, preventing electrical

current from reaching the hydraulic pump. Once airborne, the strut becomes fully extended, closing the switch that allows current to reach the hydraulic pump.

In the event of a hydraulic malfunction, the landing gear may be extended by the use of the red emergency gear extension knob. After placing the gear selector in the down position, pulling the red emergency gear extension knob releases the hydraulic pressure which is holding the gear in the up position and allows the gear to free-fall down. Emergency gear extension is limited to a maximum of 100 KIAS due to air-load on the nose gear.

Diagnosing gear abnormalities can require careful analysis. A non-illuminated gear indicator could mean the gear is not locked down, that the indicating system is malfunctioning, or simply that the bulb has burned out. (To check the bulbs, the indicator lights can be pulled out and swapped between sockets.)

When dealing with a suspected landing gear problem, it is important to verify the position of the navigation light switch. This switch will cause the green lights to be dimmed and, in some cases, make them impossible to see during the day. As with any emergency or abnormality, always refer to the appropriate checklist. Use all available information (from the gear indicators, Gear Unsafe annunciator, warning horn, and nose gear mirror) to diagnose potential system issues.



NOTE: If hydraulic pressure is lost with gear retracted, the gear will free-fall.

The nose wheel is steered through the use of the rudder pedals and is steerable 30 degrees either side of center.

Brakes

The Seminole is equipped with hydraulically actuated disk brakes on the main landing gear wheels. Braking is accomplished by depressing the tops of the rudder pedals. The hydraulic system for the brakes is independent of that for the landing gear. The brake fluid reservoir for servicing is located in the nose cone. To set the parking brake, hold the brakes and pull the black parking brake knob.



All ATP Seminoles are equipped with heavy duty brakes. Refer to the POH to make the required adjustment to performance calculations.

The parking brake is **not** to be used in training or flight checks with ATP.

Flaps

The Seminole is equipped with a manual flap system. The flaps are extended with a lever located between the two pilot seats. Flap settings are 0°, 10°, 25° and 40° and are spring-loaded to return to the 0° position.

ATP operations require Flaps 25° for all landings except the short field, which requires Flaps 40°. Flaps 25° causes a gear warning indication if the gear is not down.

Vacuum Pumps

Seminoles with conventional flight instruments are equipped with two engine-driven vacuum pumps. The vacuum system operates the attitude gyro and, on aircraft without slaving mechanisms installed, the HSI. Suction limits are 4.8 to 5.2 inches of mercury at 2000 RPM. The failure of a vacuum pump is indicated by an annunciator panel light and a red, pump inoperative indicator on the vacuum gauge. In most circumstances, the failure of one pump alone will not cause the loss of any instruments because the remaining pump should handle the entire vacuum demand.

Pitot Static

The heated pitot mast (combined pitot tube and static port) is located underneath the left wing. An alternate static source is located inside the cabin under the left side of the instrument panel for use in the event of static port blockage. When using the alternate static source, the storm window and cabin vents must be closed, and the heater and defroster must be turned on. This will reduce the pressure differential between the cockpit and the atmosphere, reducing pitot static error. The pitot static instruments are the airspeed indicator, altimeter, and VSI.

Fuel System

The Seminole, which uses 100 low lead avgas (blue), is equipped with two 55-gallon bladder nacelle tanks. One gallon is unusable in each tank.

There are two engine-driven and two electrically driven fuel pumps. The electric fuel pumps are used for engine start, takeoff, landing, and fuel selector changes.

ATP uses the electric fuel pumps for in-flight maneuvers, except for steep turns.

The aircraft is equipped with a three-position fuel selector for each engine. The positions are “ON”, “OFF”, and “X-FEED” (cross feed). The fuel selectors remain in the “ON” position during normal operations, and each engine draws fuel from the tank on the same side as the engine. When “X-FEED” is selected, the engine draws fuel from the tank on the opposite side. Fuel cannot be transferred from tank to tank. Crossfeed operation is limited to straight and level flight only.

The correct procedure for crossfeed operations to supply the left engine with fuel from the right tank is:

1. Left engine electric boost pump on.
2. Left fuel selector selected to "X-FEED."
3. Check left fuel pressure (or fuel flow, in G1000-equipped aircraft).
4. Left engine electric boost pump off.
5. Check fuel pressure (or fuel flow, in G1000-equipped aircraft).

Electrical System

Non-G1000 Seminoles are equipped with a 14-volt electrical system which utilizes push-pull type circuit breakers; a 12-volt, lead-acid battery; and two 70-amp, engine-driven alternators. Voltage regulators maintain constant 14-volt output from each alternator at varying engine RPMs, effectively sharing the electrical load. Loss of one alternator is indicated by an annunciator light and a zero indication on the loadmeter. The remaining alternator will normally provide adequate electrical power.

The electrical system on G1000 Seminoles is similar, except that the system voltage is 28 volts, the battery is 24 volts, and the alternators generate 65 amps. Loss of an alternator is indicated by an ALTR FAIL CAS warning and an ALTR AMPS indication of zero. In addition to the main battery, G1000 Seminoles have a 24V emergency battery that will automatically power the Emergency Bus when main system voltage drops below 20V. This maintains power to the PFD, COM/NAV 1, engine instruments, standby flight instrument, and audio panel for approximately 30 minutes.

An over-voltage relay in each alternator circuit provides system protection by taking an alternator offline if its output exceeds 17-volts (for 14V aircraft) or 32 volts (for 28V aircraft). If this occurs, the ALT annunciator or ALTR FAIL CAS warning will illuminate. The battery is used as a source of emergency electrical power and for engine starts. High drain items include the lights, vent fan, heater, gear hydraulic pump, radios, and PFD/MFD (if applicable). If an electrical problem arises, always check circuit breakers. If a circuit breaker is popped, reset only one time.

Heater

Heat to the cabin is supplied by a Janitrol gas combustion heater located in the nose compartment. Air from the heater is distributed by a manifold to the ducts along the cabin floor to outlets at each seat and to the defroster outlet. Operation of the heater is controlled by a three-position switch located on the instrument panel labeled "CABIN HEAT", "OFF", and "FAN." Airflow and temperature are regulated by the three levers to the right of the switch: "AIR INTAKE", "TEMP", and "DEF."

For cabin heat, the "AIR INTAKE" lever must be fully open and the "CABIN HEAT" switch on. This simultaneously starts fuel flow and ignites heater. During ground

operation, it also activates the ventilation blower. When cabin air reaches the temperature selected on the “TEMP” lever, ignition of the heater cycles automatically to maintain the selected temperature.

Two safety switches located on the front of the heater unit prevent both fan and heater operation when the air intake lever is in the closed position. When the heater is on, a ventilation blower actuates any time the landing gear is extended. In flight, a micro switch, which actuates when the gear is retracted, turns off the ventilation blower so that the cabin air is circulated by ram air pressure only.

An overheat switch in the heater acts as a safety device to render the heater inoperative if a malfunction should occur. Should the switch deactivate the heater, the red “HEATER OVER TEMP” annunciator light on the instrument panel will illuminate (in G1000-equipped aircraft, a CAS warning will appear reading HTR OVRHEAT). The overheat switch is located on the aft inboard end of the heater vent jacket. A red reset button is located on the heater shroud in the nose cone compartment.

To prevent activation of the overheat switch upon normal heater shutdown during ground operation, turn the three-position switch to “FAN” for two minutes with the air intake lever in the open position before turning the switch off. During flight, leave the air intake open for a minimum of 15 seconds after turning the switch to off.

Fuel is supplied to the heater at a rate of ½ gallon per hour from downstream of the left fuel selector and filter.

To introduce outside, unheated air into the cabin during flight, the “AIR INTAKE” lever should be open and the “CABIN HEAT” off. A fresh air blower is installed to provide airflow during ground operation. It is operated by a high/low blower fan switch.

Stall Warning Horn

The Seminole is equipped with two electric stall detectors located on the left wing. The inboard detector provides stall warning at flaps 25° or 40° and the outboard at flaps 0° or 10°. The purpose of the two tabs is to provide adequate stall warning at varied angles of attack. The electric stall tabs are deactivated on the ground through the use of the squat switch on the left (1979 model) or right (2000 and later models) main landing gear. In G1000 Seminoles, the warning is an aural annunciation ("Stall... Stall... Stall..."), while in conventional or G500 Seminoles it is a continuous tone.

Emergency Exit

The emergency exit is the pilot’s left side window. Use this when emergency egress becomes necessary on the ground and the main entry door is unavailable due to fire, etc. The emergency exit release handle is located beneath the thermoplastic cover on the vertical post between the first and second left side

windows. To exit the aircraft, remove the thermoplastic cover, pull the release handle forward and then push the window out. The window then will fall free from the fuselage.

Inoperative Instruments & Equipment per FAR 91.213

ATP's aircraft do not operate under the guidance of a minimum equipment list (MEL), and instead operate in accordance with the following FAR 91.213 subpart. Because this is only an excerpt, the complete subpart should be referred to if necessary:

> (3) The inoperative instruments and equipment are --
>
> > (i) Removed from the aircraft, the cockpit control placarded, and the maintenance recorded in accordance with §43.9 of this chapter; or
> >
> > (ii) Deactivated and placarded "Inoperative." If deactivation of the inoperative instrument or equipment involves maintenance, it must be accomplished and recorded in accordance with part 43 of this chapter;
>
> (4) A determination is made by a pilot, who is certificated and appropriately rated under part 61 of this chapter, or by a person, who is certificated and appropriately rated to perform maintenance on the aircraft, that the inoperative instrument or equipment does not constitute a hazard to the aircraft.

Lynx NGT-9000

Piper Seminoles equipped with the Lynx NGT-9000 must use the Lynx as the Primary Traffic information source. The standard avionics configuration for the Lynx consists of the homepage with transponder information on the left side of the screen and traffic information on the right.



Garmin G500

2012 and 2013 model Piper Seminoles are equipped with the Garmin G500 electronic flight deck.

PRIMARY FLIGHT DISPLAY (PFD)



MULTI-FUNCTION DISPLAY (MFD)

G500 Differences

- The G500 powers on and off with the battery master, not the avionics master.
- There are no vacuum driven instruments, vacuum pumps, or suction gauge. Instruments are electrically powered.
- The standby attitude indicator must be erected on power-up by pulling the cage knob.
- Test the standby attitude battery as directed by the After Start Checklist by pressing and holding the “STBY PWR” button until amber LED begins flashing. A continuous green light beneath “TEST” during the full sequence indicates that the battery is good. A red light illuminated anytime during the test indicates that the battery is not charged and may require replacement.
- The standby attitude indicator warning light will blink for approximately 1 minute when the battery master switch is turned off. DO NOT press the STBY PWR button as it causes continuous operation of the gyro on its emergency battery. Exit the airplane with either a blinking LED or red flag.
- G500 equipped Seminoles do not have a conventional turn coordinator. A slip-skid indicator is located at the top of the attitude indicator. Step on the “brick” instead of the “ball”. Use the reference lines and the magenta line that appears above the heading indicator to identify a standard rate or half-standard rate turn.
- Outside air temperature (OAT) displays on PFD under the airspeed tape. There is no OAT probe or analog gauge between the windows.
- Ground track can be identified on the heading indicator by a small magenta diamond near the lubber line (only visible when ground track is different than heading).

- The digital altitude and airspeed readouts are very sensitive and can cause some pilots to continuously make corrections for insignificant deviations. Pilot action - Don't overcorrect for deviations of a few feet. Crosscheck digital and analog standby instruments to avoid the tendency to overcorrect.

Refer to the complete G500 Pilot's Guide located in ForeFlight Documents and in the Student Extranet Library.

G500 Components

The G500 is comprised of 6 main components:

- Primary Flight Display (PFD, left) and Multi-Function Display (MFD, right)
- Attitude Heading Reference System (AHRS)
- Air Data Computer (ADC)
- Magnetometer
- Temperature Probe
- Dual Garmin GNS 430 GPS

The PFD (left) shows primary flight information in place of traditional pitot-static and gyroscopic instruments and also provides an HSI for navigation. ATP procedures call for configuring the MFD (right) to display traffic information service (TIS).

The Attitude Heading Reference System (AHRS) contains tilt sensors, accelerometers, and rate sensors to provide attitude and heading information on the PFD.

The Air Data Computer (ADC) compiles information from the pitot-static system and an outside air temperature sensor to provide pressure altitude, airspeed, vertical speed, and outside air temperature on the PFD.

The magnetometer senses the earth's magnetic field and sends data to the AHRS for processing to determine magnetic heading.

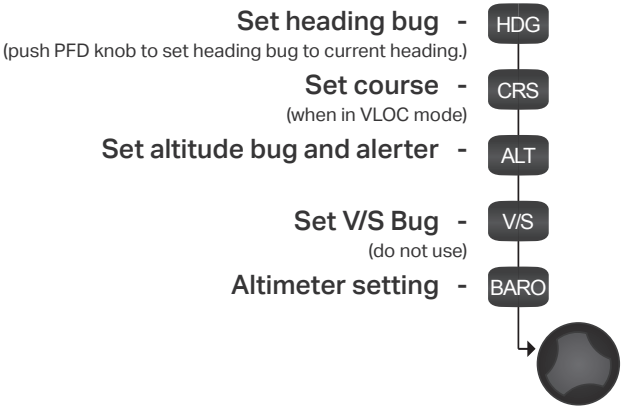
The temperature probe provides outside air temperature (OAT) data to the ADC.

The dual Garmin GNS 430 GPSs provide input to the AHRS and PFD/MFD.

CAUTION: The GNS 430 and G500 units each have their own databases. Navigation, terrain and map information on the G500 Multi-Function Display (MFD) may not be current and is not to be used for navigation. Use the G500 MFD for traffic information.

G500 PFD Functions

These buttons toggle the function of the PFD knob.



CDI needle indicates NAV source: green for VLOC / magenta for GPS.

G500 Standby Instruments

Standby Attitude Indicator

An electrical standby attitude indicator is powered and charged by the electrical system during normal operations. During an electrical failure, the standby attitude indicator will continue operating from its internal dedicated battery for approximately 60 seconds while and an amber blinking LED prompts you to press the “STBY PWR” button, which continues operation until the internal battery is depleted. A fully charged standby attitude indicator battery should provide up to 1 hour of normal standby attitude indicator operation; however, a situation requiring use of the standby attitude indicator is an emergency. Exit IMC and land as soon as possible.

If the pilot fails to activate the standby battery within 60 seconds, the STBY PWR button can still be pressed to activate the standby battery. However, the gyro may not be oriented to the airplane depending on the time elapsed prior to pressing the STBY PWR button. Press the STBY PWR button while the amber LED is blinking for best results.

Standby Attitude Indicator Battery Check

Prior to flight, the electric standby indicator battery must be tested. In order to accomplish the test, aircraft electrical power must be on (Battery Master ON) and the standby attitude indicator gyro must be fully operational with the red gyro flag out of view. Test the standby battery as called for by the After Start Checklist.

- Press and hold the "STBY PWR" button for approximately four seconds. This puts the gyro in a one-minute battery test mode. The amber LED next to STBY PWR will flash during the test sequence.
- A continuous green light illuminated beneath "TEST" during the full sequence indicates that the standby battery is good.
- A red light illuminated anytime during the test indicates that the battery is not charged and may require replacement. Contact maintenance.

Standby Attitude Indicator Shutdown

During a normal shutdown following the Shutdown Terminate Checklist, do not press the "STBY PWR" button. Doing so activates emergency operation and depletes the internal standby battery. Turn the battery master switch OFF when prompted by the checklist. Allow the amber LED to blink for approximately 60 seconds, followed by the appearance of the red gyro warning flag, which indicates the unit is powered down.

Standby Altimeter & Airspeed Indicator

The pitot-static system provides information to the standby altimeter and airspeed indicator.

G500 Failures & Partial-Panel Approaches

For partial panel training and checkrides, the two most common training scenarios are simulated AHRS failure and PFD failures.

Failure Condition	Simulated By	Instrument Approaches Available
AHRS Failure	Cover Attitude Indicator (ADI)	All precision and non-precision
PFD Failure	Dim PFD/MFD screens	Only GPS approach
Electrical Failure	No simulated failure available	None
ADC Failure	No simulated failure available	All precision and non-precision

Circuit breaker-simulated failures are prohibited in ATP aircraft. Piper and Garmin advise against pulling circuit breakers as a means of simulating failures on the Garmin G500 system. Pulling circuit breakers–or using them as switches–has the potential to weaken the circuit breaker to a point at which it may not perform its intended function. Also reference Advisory Circulars 120-80, 23-17B, and 43.13-1B.

Attitude and Heading Reference System (AHRS) Failure

Indications:

1. The sky/ground presentation is removed.
2. A red X appears across the Attitude Direction Indicator (ADI).
3. Yellow “ATTITUDE FAIL” and “HDG” alert messages appear on the PFD.
4. A “TRK” message appears to the right of the ground track at the top of the compass rose.
5. Rate-of-turn information is unavailable.
6. “HDG LOST”, “HDG FAULT”, and "TRK TRAFFIC” alert messages appear on the MFD.



AHRS Failure

The PFD continues displaying airspeed, altitude, vertical speed, compass rose and ground track. Ground track and compass rose indications are supplied by GPS, indicated by a “TRK” message. Any precision or non-precision approach is available using the HSI on the PFD.

Pilot Action

1. Use standby attitude indicator.
2. Continue using HSI on PFD. Verify track against magnetic compass heading.
3. Precision (ILS) and non-precision (GPS, Localizer, and VOR) approaches can be accomplished.

PFD Failure

Indications

1. PFD Screen is dark.

Pilot Action

1. Refer to the standby instruments.

2. Use the GPS CDI page for navigation and approaches.
   CLR - Press and hold for 3 seconds to return to default CDI page.
3. Only GPS non-precision approaches can be accomplished.

During an MFD failure, with the PFD functioning normally, all approaches are available for use.

Electrical Failure

Indication

1. The G500 and GPS systems will be inoperative / dark.
2. The “STBY PWR” button on the standby attitude indicator will begin blinking.

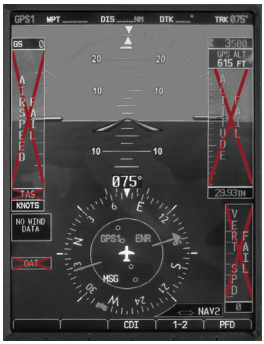
Pilot Action

1. Use standby attitude indicator. Press the "STBY PWR" button right of the blinking LED to continue operating using its internal battery.
2. Use standby airspeed, altimeter and compass.
3. Declare an emergency and exit IMC as soon as practicable. The manufacturer does not specify the endurance time of the integral emergency battery.

Air Data Computer (ADC) Failure

Indications

1. Loss of data accompanied by a red X and yellow alert messages occurs over:
   - Airspeed
   - Altitude
   - Vertical speed
   - True airspeed (TAS)
   - Outside air temperature (OAT)
2. Wind calculations are unavailable



ADC Failure

Attitude and heading references will function normally on the PFD.

Pilot Action

1. Use standby airspeed indicator and altimeter.

There is no backup for the VSI, but known pitch attitudes using the attitude indicator, power settings, and airspeeds produce consistent rates.

Garmin G1000

Some Piper Seminoles are equipped with the Garmin G1000 electronic flight deck.

G1000 Components

The G1000 is comprised of several main components, called Line Replaceable Units (LRUs):

- Primary Flight Display (PFD)
- Multi Function Display (MFD)
- Integrated Avionics Units (2)
- Attitude and Heading Reference System (AHRS)
- Air Data Computer (ADC)
- Engine/Airframe Unit
- Magnetometer
- Audio Panel
- Transponder

The PFD (left screen) shows primary flight information in place of traditional pitot-static and gyroscopic instruments, and also provides an HSI for navigation.

The MFD (right screen) provides a GPS-enabled moving map with traffic and weather information. It can also be used to display waypoint/airport information, flight plans, instrument procedures, trip planning utilities, and system setup/configuration information.

The Integrated Avionics Units each contain a WAAS-enabled GPS receiver, a VHF nav/comm radio, and a flight director. They also serve as communications hubs to relay information from the other LRUs to the PFD and MFD. For redundancy, one IAU is connected to each display, and they do not communicate with each other directly.

The Attitude and Heading Reference System uses accelerometers and rate sensors, along with magnetic field readings from the magnetometer and GPS information from the IAUs, to provide aircraft attitude and heading information to the flight displays and IAUs.

The Air Data Computer processes data from the pitot/static system as well as the OAT probe to provide pressure altitude, airspeed, vertical speed, and air temperature data to the system.

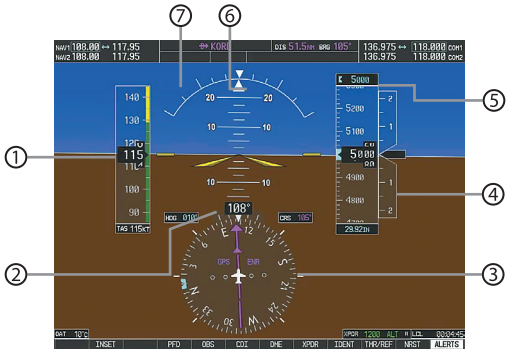
The Engine/Airframe Unit receives and processes signals from the engine and airframe sensors (engine RPM and temperatures, fuel quantity, gear position, etc.).

The magnetometer measures the local magnetic field and sends data to the AHRS to determine the aircraft's magnetic heading.

The audio panel is installed between the two display screens and integrates controls for the nav/com audio, intercom system, and marker beacon receiver. It also controls manual display reversionary mode (which can shift the primary flight instruments to the MFD).

The transponder is a Mode S device, controlled via the PFD, that also provides ADS-B In/Out capability.

G1000 Flight Instruments



Primary Flight Display (Default)

① Airspeed Indicator
② Turn Rate Indicator
③ Horizontal Situation Indicator (HSI)
④ Vertical Speed Indicator (VSI)
⑤ Altimeter
⑥ Slip/Skid Indicator
⑦ Attitude Indicator

The G1000 PFD displays the same flight information as the conventional "six-pack", but pilots should be aware of the following considerations.

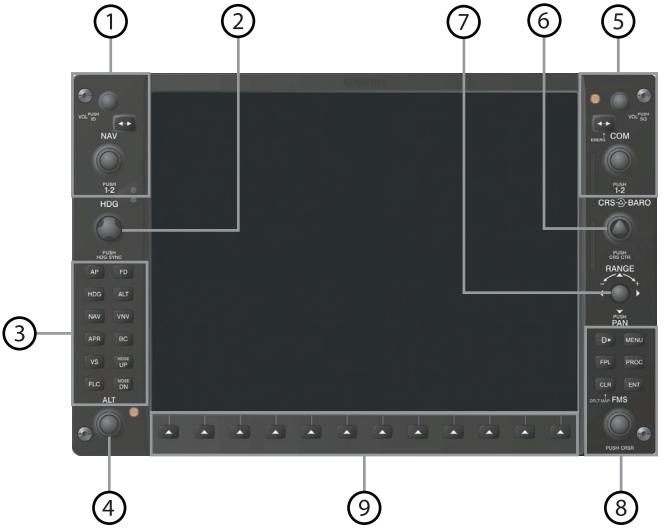
Airspeed and altitude information are displayed with moving tapes and a digital readout of the current airspeed and altitude to the nearest knot / 20 feet, respectively. This precision leads some pilots to overcontrol the aircraft, continuously making corrections for insignificant deviations. Be sure not to overcorrect for deviations of a few feet or knots.

The information traditionally displayed on the turn coordinator is split between two locations on the screen. The inclinometer ("ball") is replaced with a white "brick" under the pointer at the top of the attitude indicator. "Step on the brick" to center it and maintain coordinated flight. The rate of turn indication is provided by a magenta trend vector at the top of the HSI. Tick marks are provided for half-standard and standard rate turns.

On the HSI, a small magenta diamond indicates the aircraft's current ground track. (This diamond may not be visible if crosswinds are minimal and the track is nearly equal to the heading.) Also, pilots should note the color of the CDI needle to determine the current navigation source. Magenta needles indicate GPS, while green needles indicate VOR or LOC.

G1000 Controls

The G1000 has duplicate sets of controls on the PFD and MFD bezels. Using the controls towards the center of the aircraft (on the right side of the PFD and the left side of the MFD) helps to ensure that both student and instructor can see each other's inputs.



PFD/MFD Controls

Left Side - Top to Bottom

① NAV Radio Controls: Use the NAV knob, along with the frequency transfer key, to tune NAV receiver frequencies. Turn the VOL knob to control the volume, and press the knob to toggle the Morse code identifier on/off.

② HDG Knob: Sets the heading bug on the HSI.

③ AFCS Keys: Used to program the Garmin GFC 700 Automatic Flight Control System. (Not installed on all aircraft.)

④ ALT Knob: Sets the altitude bug on the altimeter.

Right Side - Top to Bottom

⑤ COM Radio Controls: Use the COM knob, along with the frequency transfer key, to tune COM receiver frequencies. Turn the VOL knob to control the volume, or press to turn the automatic squelch on or off.

⑥ CRS/BARO Knobs: Turn the outer, large knob to set the barometric pressure setting for the altimeter. Turn the small, inner knob to select a course on the HSI when in VOR or OBS mode.

⑦ RANGE Joystick: Turn to adjust map range. Press to activate the map pointer.

⑧ FMS Keys/Knob: Use these to program flight plans, enter waypoints, select instrument procedures, etc.

Bottom Edge

⑨ Softkeys: There are 12 softkeys along the bottom edge of each display with functions that vary depending on context.

ATP Review the G1000 Pilot's Guide for your airplane for more information on the G1000's features. These are available in the ATP Library and in ForeFlight Documents.

Standby Instruments

G1000-equipped Piper Seminoles are also equipped with a standby flight instrument (either an Aspen Evolution Backup Display or a Garmin G5) mounted on the left side of the instrument panel. Both Garmin and Aspen units are fully digital, independent flight instrument displays providing attitude, barometric altitude, airspeed, heading, vertical speed, slip/skid, and turn rate indications. Should the G1000 fail (say, due to a failed display screen or AHRS), the pilot can control the aircraft using the standby instrument. If this occurs, exit and avoid IFR conditions as soon as practical.

ATP G1000-equipped Seminoles do not have traditional magnetic compasses. Instead, the standby instruments provide a backup source of magnetic heading information. The Aspen EBD contains an internal magnetometer, while the Garmin G5 is accompanied by an external GMU 11 magnetometer unit.

The standby instrument can draw electrical power from a few different sources, to ensure the pilot has uninterrupted access to critical flight information:

- In normal operation, the standby instrument is powered by the engine-driven alternator via the Essential Bus.
- If the alternator fails, the primary battery will continue to power the standby instrument via the Essential Bus.

- Once the primary battery is depleted (or if the primary electrical system otherwise fails), the Seminole's emergency battery will power the standby instrument via the Emergency Bus for approximately 30 minutes.
- If all electrical power to the standby instrument is removed, a backup internal battery in the unit will continue to power it for at least 30 minutes in the case of the Aspen EBD, or up to 4 hours in the case of the Garmin G5.

Prior to flight, the pilot must check the power level of the emergency battery using the procedures in the ATP Seminole Checklists. With the alternator and battery master off, move the emergency battery switch to the ARM position. The PFD and standby instrument should both power on with all instrument indications functional, and the E VOLTS reading on the PFD must be at least 23.3 volts. If E VOLTS is low prior to engine start, check again following run up.

Activation of the emergency battery following electrical failure indicates an emergency situation. Land as soon as possible.

The standby instrument will not automatically switch to its internal battery if the aircraft is on the ground (inferred from low airspeed). This ensures the unit does not remain on after aircraft shutdown. However, if the Aspen EBD does not automatically turn off during aircraft shutdown, press and hold the red REV button on the upper right for up to 20 seconds until the "Shutdown in progress" message appears. The Garmin G5 can be turned off by pressing and holding the power button for five seconds.

G1000 Failures & Partial-Panel Approaches

Training Considerations

For partial-panel training and checkrides, the two most common training scenarios are PFD failures and ADAHRS failures.

- **PFD Failure:** Simulate by dimming the PFD screen. The student should respond by pushing the DISPLAY BACKUP button to activate reversionary mode and move the flight instrument displays to the MFD. All instrument procedures remain available. Use the Inset Map for situational awareness.
- **ADC or AHRS Failure:** The ADC and AHRS have various failure modes that can cause one or more instrument indications to become unavailable. To simulate a worst-case scenario in which all of the G1000's flight instruments are unusable, dim the PFD screen and do **not** activate reversionary mode. Then, fly the airplane using the standby instruments. The MFD should remain on the moving map screen for situational awareness. GPS approach procedures remain available. Set the MFD fields to TRK, DTK, XTK, and DIS to maintain situational awareness of your position relative to the intended track (in lieu of the CDI).

Other failure modes in which some (but not all) instruments are unavailable can be simulated using paper or foam cutouts that hang from the COM and NAV knobs and cover up particular areas of the PFD screen. ATP does not provide these cutouts.

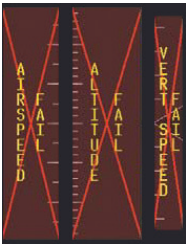
The simulation of failures by pulling circuit breakers is prohibited in ATP aircraft. Piper, Garmin, and the FAA all advise against pulling circuit breakers as a means of simulating failures on the G1000 system. Pulling circuit breakers, or using them as switches, has the potential to weaken the circuit breaker to a point at which it may not perform its intended function.

LRU Failures

If an LRU or an LRU function fails, a red or amber X is displayed over the window(s) corresponding to the failed data. If this occurs, follow the appropriate emergency checklist. Generally, this involves checking the circuit breaker for the affected LRU, then (if the problem is not fixed by resetting the breaker) using the standby instruments to exit IFR conditions and land as soon as practical.



AHRS Failure



ADC Failure

AHRS Modes

The AHRS uses GPS, magnetometer, and air data to assist in attitude/heading calculations, in addition to the data from its internal sensors. Loss of this external data can affect the availability of attitude and heading information, even if the AHRS itself is functional. Either GPS or air data must be available for the AHRS to provide attitude information. Additionally, loss of magnetometer data will result in invalid heading information.

If the AHRS cannot provide valid heading information, the course pointer on the HSI will point straight up, effectively converting it into a standard, fixed-card course deviation indicator. As a result, pilots can still perform partial-panel instrument approach procedures following an AHRS failure. Cross-reference between heading information from the magnetic compass and course information from the PFD.

Display Failures

If either display fails, the G1000 should automatically enter reversionary mode, in which important flight information is presented in a condensed format on the remaining display(s). Reversionary mode can also be activated manually by pressing the red DISPLAY BACKUP button on the audio panel. Engine Indication System readings appear on the left edge of the screen, and the inset map appears at lower right.



Reversionary Mode (Failed PFD)

Because the IAUs are not cross-linked, any functions handled by just one IAU will be lost if its corresponding display fails. If the PFD fails, NAV1, COM1, and GPS1 will be unavailable. If the MFD fails, NAV2, COM2, and GPS2 are unavailable. Other optional avionics may also become unavailable, depending on the particular avionics configuration.

Electrical Failure

If the alternator fails, a red "ALTR FAIL" CAS warning message will appear on the PFD. All G1000 equipment will initially remain on, powered by the main battery. Follow the appropriate emergency checklist to verify the failure and attempt to reset the alternator.

If the alternator is still failed, continue to the electrical load shedding portion of the checklist. Part of this procedure is to turn off the avionics master. This will disable the MFD and #2 IAU, along with optional avionics. The PFD will remain active and switch to reversionary mode.

Once the main battery is depleted, the emergency battery will activate and provide approximately 30 minutes of power to the emergency bus. This will maintain power to the PFD, but the transponder and the cooling fans will shut off. Declare an emergency, exit IMC, and land as soon as possible.

Garmin G5

Some 1979-model Piper Seminoles are equipped with two Garmin G5 Electronic Flight Instruments in place of the vacuum-powered attitude indicator and horizontal situation indicator. Aircraft with G5 avionics have had their vacuum system equipment removed (including related gauges and annunciator lights).

G5 Pages

Each G5 has two pages: the PFD Page and the HSI Page. During installation, the units are configured to display the PFD Page on the top unit and the HSI Page on the bottom unit. The PFD displays attitude, airspeed, altitude, turn rate, heading, and vertical speed information, using a similar layout to the Garmin G1000 (but in a smaller form factor). To maintain commonality with aircraft with all-conventional gauges, standard procedure is to use the G5 PFD as the primary source of attitude information only, while using the conventional gauges for airspeed, altitude, turn rate, and vertical speed information.

Slight mismatches of a few knots or tens of feet between the airspeed/altitude information on the PFD and the conventional instruments are common and not cause for concern. When setting the altimeters, enter the same pressure setting on both; do not modify one setting just to make the altimeters match.

The HSI Page functions like a standard HSI, showing heading and either VOR/LOC or GPS course information depending on which is selected on the NAV 1 GNS 430. The Selection Knob normally adjusts the heading bug; to select a course, first press the knob to enter the menu and select “Course” (or “OBS” if using GPS data), then use the knob to select the desired course.

G5 Failures

As with other Garmin glass-panel instruments, failures of G5 instrument functions are indicated by a red “X” over the affected data. Failures in flight can be addressed by using other cockpit instruments or (in the case of attitude failure) by switching the bottom unit to show the PFD page. To switch pages, press the Selection Knob to activate the unit’s menu, then rotate to select the PFD Page. An [!] icon in the lower left indicates a new system message. Press the Selection Knob and rotate to select and read the message. The G5 Pilot's Guide lists possible messages and their meanings (most of which alert the pilot to system faults).

G5 Electrical Power

For redundancy, the upper G5 unit is powered from the main bus and will turn on with the battery master switch, while the lower one is powered from the avionics bus and will turn on with the avionics master. The G5 instruments also have internal backup batteries that can power the device for up to 4 hours if aircraft electrical power is lost. If this occurs, a system message will appear and the battery status indicator in the upper left of the screen will display the remaining battery charge time. Exit IMC and land as soon as practicable.

Standard Avionics Configurations

Automatic zoom must be disabled. Set ranges to view approximately a 2 NM radius in the traffic pattern or congested areas; a 6 NM radius for departure, arrival, and practice area operations; and a 12 NM radius for enroute VFR or IFR.

Garmin G1000

Taxiing

MFD: Map page with range set to airport diagram view.

⚠ Airport diagram may be expired, reference for situational awareness only.



Traffic Pattern Operations

PFD: Traffic map (INSET left).

MFD: Map page with traffic information active.

- ***NXi:*** Detail set to Detail-3
- ***Other:*** Declutter set to DCLTR-1

Enroute and Airwork

PFD: Active with appropriate nav source (needles) active.

MFD: Map page with traffic information active. Traffic inset should be included on the PFD for added traffic awareness.

- ***NXi:*** Detail set to Detail-3
- ***Other:*** Declutter set to DCLTR-1

Full Panel Approaches

PFD: Active NAV source on HSI/Traffic map (INSET left).

MFD: Map page with traffic information active and orientation set to track up.

- ***NXi:*** Detail set to Detail-3
- ***Other:*** Declutter set to DCLTR-1



G1000 Standard Configuration

Partial Panel Approaches

PFD: Dimmed or covered.

MFD: Reversionary Mode.

Map Overlay: On with Traffic Information active.



G1000 Partial Panel Configuration

Pilots can exercise PIC judgment to briefly switch to other pages or settings with information helpful to the safe, efficient conduct of the flight. After doing so, promptly switch back to the standard configurations.

Garmin G500 w/ Dual GNS430

Taxi & Traffic Pattern Operations

MFD: Map page group, traffic map page (map page 3).

MFD databases are not regularly updated in G500-equipped aircraft. Use GNS430 for tasks requiring current database information (geographical location, airspace, airports and navaids, etc.)

COM/NAV 1: Nav page group, Moving map page (Nav page 2).

COM/NAV 2: Nav page group, Traffic information page (Nav page 3).



Dual Garmin GNS 430 GPS Configuration

Enroute, Airwork, and Full Panel Approaches

PFD: Active with appropriate nav source active on HSI.

MFD: Map page group, traffic Map page (Map page 3).

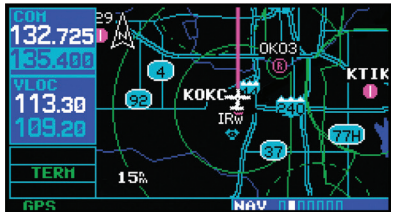
COM/NAV 1: Default Nav page (Nav page 1), page values set to DIS, GS, DTK, TRK, XTK, ETE.

VLOC Button: Selected to appropriate nav source.

COM/NAV 2: Moving Map page (Nav page 2), with traffic information active and orientation set to TRACK UP.







G500 w/ dual 430s - Enroute

Partial Panel Approaches:

PFD: Dimmed or covered. Use standby instruments.

MFD: If not dimmed, map page with traffic information active and orientation set to track up.

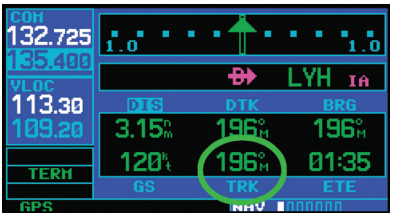
COM/NAV 1: Approach loaded, Default Nav page (Nav 1).

Default Nav page values: DIS, GS, DTK, TRK, XTK, ETE

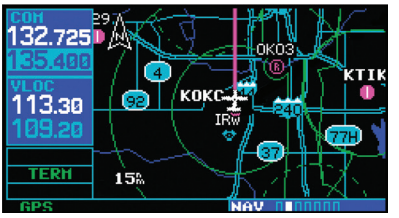
COM/NAV 2: Nav page group, Moving map page (Nav page 2), with traffic information active and orientation set to TRACK UP.

VLOC Button: Selected to appropriate nav source.

Course Guidance: TRK information (on GPS1).



GPS Approach with No HSI: Use GPS1 set to CDI page



GPS2 Set to Moving Map Page (Nav 2)

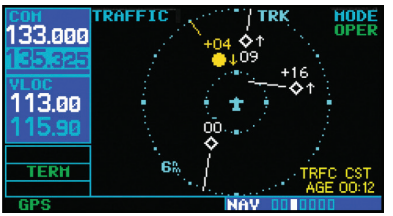
For PFD failures on the G500, the only approach option is a GPS approach, using the GPS CDI for course guidance.

Single Garmin GNS430

Taxi & Traffic Pattern Operations

COM/NAV 1: Traffic Information Page (Nav page 3)

NGT-9000: Home page with traffic information active

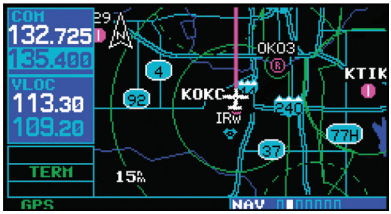




Enroute, Airwork & Full Panel Approaches

COM/NAV 1: Moving map page (Nav page 2), orientation set to track up, traffic information active

NGT-9000: Home page with traffic information active





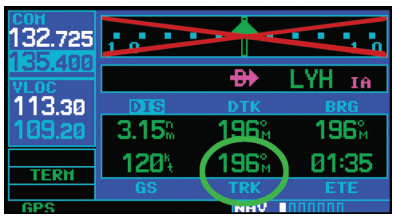
Partial Panel Approaches

COM/NAV 1: Default Nav page (Nav page 1), page values set to DIS, GS, DTK, TRK, XTK, ETE

NGT-9000: Home page with traffic information active

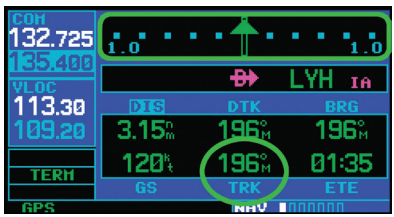
Attitude and heading indicator covered





Partial Panel VOR / ILS Approaches - Use Nav 2 OBS





Partial Panel GPS Approach Configuration

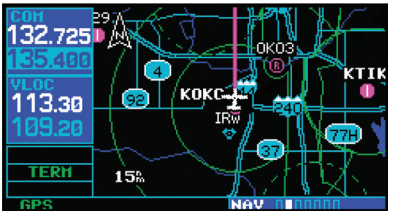
Dual Garmin GNS430 (with or without Lynx NGT-9000)

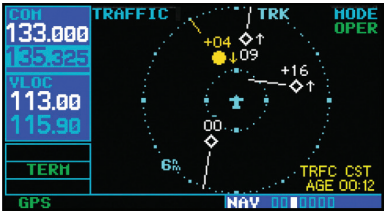
Taxi & Traffic Pattern Operations

COM/NAV 1: Nav page group, Moving map page (Nav page 2).

COM/NAV 2: Nav page group, Traffic information page (Nav page 3).

NGT-9000: Home page with transponder info on left side and traffic on right side







Enroute, Airwork & Full Panel Approaches

COM/NAV 1: Nav page group, Moving map page (Nav page 2), orientation set to track up

COM/NAV 2: Nav page group, Traffic information page (Nav page 3)

NGT-9000: Home page with Traffic

Partial Panel Approaches

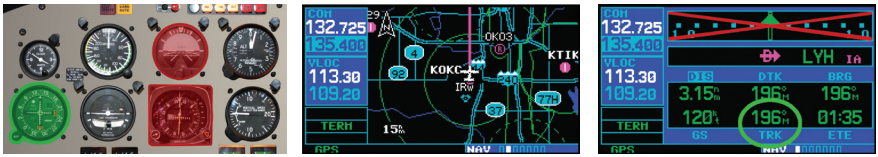
COM/NAV 1: Nav page group, Moving map page (Nav page 2), orientation set to track up

COM/NAV 2: Default Nav page (Nav page 1)

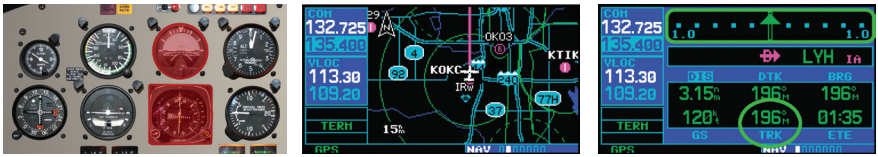
Default Nav page values: DIS, GS, DTK, TRK, XTK, ETE

NGT-9000: Home page with Traffic

Attitude and heading indicator covered



Partial Panel VOR / ILS Approaches - Use Nav 2 OBS



Partial Panel GPS Approach Configuration

GPS Setup for VFR Airwork & Maneuvers

When in a familiar area performing VFR maneuver training, or when remaining in the traffic pattern for takeoff/landing practice, at least one GPS unit should be selected to the Traffic page. If two units are present, one may be selected to the Moving Map page. To avoid redundant information, never have two systems on the same page.

Aircraft maneuvering will cause errors in the display. These errors primarily affect relative bearing information and traffic target track vector (it will lag). Traffic information is provided as an aid in visually acquiring traffic. It is the responsibility of the pilot to see and maneuver to avoid traffic.

List of Pilots Guides

- G500 Pilot's Guide
- GNS430 Pilot's Guide
- Lynx NGT-9000 Pilot's Guide
- G1000 Pilot's Guide
- G5 Pilot's Guide

Notable Differences Between Seminoles by Model Year

	1979-80	2000-19	2022-Later
Engines	Left: O-360-E1A6D Right: LO-360-E1A6D	Left: O-360-A1H6 Right: LO-360-A1H6	Left: IO-360-B1G6 Right: LIO-360-B1G6
Oil Capacity	• 4 to 6 Quarts • ATP Minimum 4.5 Quarts	• 6 to 8 Quarts • ATP Minimum 6.5 Quarts	
Minimum Oil Pressure	15 PSI	25 PSI	
Landing Gear Mute Switch	N/A	Do not use.	
Primer	• Manual/Plunger • Ports Fuel Into Cylinders 1, 2 & 4	• Button/Electric • Electric Fuel Pump Must be On • Press Primer Button 2-3 Sec. before Starting Cold Engine • Ports Fuel Into Cylinders 1, 2 & 4	• N/A: priming achieved by advancing mixture to RICH with fuel pump on
Recognition Lights	N/A	• Located on Each Wingtip • Used Along With Landing Light for Recognition	
Fin Strobe & Strobe Light Rocker Switch	Anti-Collision Lights Only (Strobes)	• Rocker Switch Down for Ground Use (Fin Strobe / Rotating Beacon) • Rocker Switch Up Before Takeoff / Flight Use (Strobes)	
"LO BUS" Annunciator Light	N/A	Illuminates Red when Alternators not Providing 14 V DC to the tie bus, and bus voltage drops to battery voltage, 12.5 V DC	N/A
Unfeathering Accumulators	N/A	• Stores Engine Oil Under Pressure for Air-Starting the Engine • Releases Stored Pressure Back to Prop Governor when Prop Lever Moved Forward • Oil Pressure Drives the Blades from the Feathered Position Toward Low-Pitch (Windmilling) • Windmilling Prop with Addition of Fuel and Ignition will Allow the Engine to Start	
Airstart	• Starter Only • Be Sure to Only Engage the Proper Side of the Rocker Start Switch • Pressing the Wrong Side will Cause Engine Damage to the Operating Engine	• Unfeathering Accumulator The 2000 Seminole utilizes a prop unfeathering accumulator which stores engine oil under pressure for airstarting the engine. The stored pressure is released back to the prop governor when the prop lever is moved forward. Oil pressure drives the blades from the feathered position toward low-pitch windmilling. When the prop is windmilling, the addition of fuel and ignition will allow the engine to start.	

Piper Seminole V-Speeds

Speeds listed below are in Knots Indicated Airspeed (KIAS).

V-Speed	KIAS	Description	Airspeed Indicator Marking
V_{SO}	55	Stall speed in landing configuration	Bottom of White Arc
V_{MC}	56	Minimum controllable airspeed	Red Line
V_S	57	Stall speed with zero flaps	Bottom of Green Arc
V_R	75	Rotation speed (start rotation)	
V_X	82	Best angle of climb	
V_{XSE}	82	Best angle of climb single-engine	
V_{SSE}	82	Safe speed for intentional engine failure	
V_Y	88	Best rate of climb	
V_{YSE}	88	Best rate of climb single-engine	Blue Line
V_{FE}	111	Maximum flap extension speed	Top of White Arc
V_{LO} (Up)	109	Maximum gear retraction speed	
V_{LO} (Down)	140	Maximum gear extension speed	
V_{LE}	140	Maximum speed with gear extended	
V_{NO}	169	Max Structural Cruising Speed	Top of Green Arc
V_{NE}	202	Never exceed speed	Red Line
V_A	135	Maneuvering speed at 3800 pounds	
V_A	112	Maneuvering speed at 2700 pounds	

Maximum demonstrated crosswind 17

Performance Charts

All performance charts will be covered by the instructor and are not limited to the following:

- Takeoff Distance to Clear a 50' Obstacle
- Accelerate Stop Distance
- Single Engine Rate of Climb
- Single Engine Service Ceiling
- Landing Distance

Sample Weight & Balance Problem

Complete the following sample weight and balance problem.

Conditions

Basic Empty Weight: ..... **2,540.5 lbs.**
(Remember to use actual aircraft BEW for flight check.)

Front Pilots: ..... **370 lbs.**

Rear Passengers: ..... **170 lbs.**

Baggage: ..... **3 Bags @ 75 lbs. and 2 Bags @ 50 lbs.**
(May need to relocate some baggage to rear passenger seats.)

Max Ramp Weight: ..... **3,816 lbs.**

Max Takeoff/Landing Weight: ..... **3,800 lbs.**

Max Baggage Weight: ..... **200 lbs.**

Max Usable Fuel: ..... **108 gal.**

		Weight	×	Arm	=	Moment
Basic Empty Weight				86.13		
Front Pilots	+			80.50	+	
Rear Passengers	+			118.10	+	
Baggage 200 lbs. Max	+			142.80	+	
Zero Fuel Weight	=			CG CG = Moment / Weight	=	
Usable Fuel	+			95.00	+	
Ramp Weight	=					
Taxi Fuel (2.65 Gal.)	–	16		95.00	–	1,520
Takeoff Weight	=			CG CG = Moment / Weight	=	
Fuel Burn	–					
Landing Weight	=			CG CG = Moment / Weight		

Calculate the Following

1. Zero Fuel CG
2. Usable Fuel
3. Takeoff Weight
4. Takeoff CG
5. From comparing the Takeoff CG and Zero Fuel CG, which direction does the CG move as fuel is burned off?

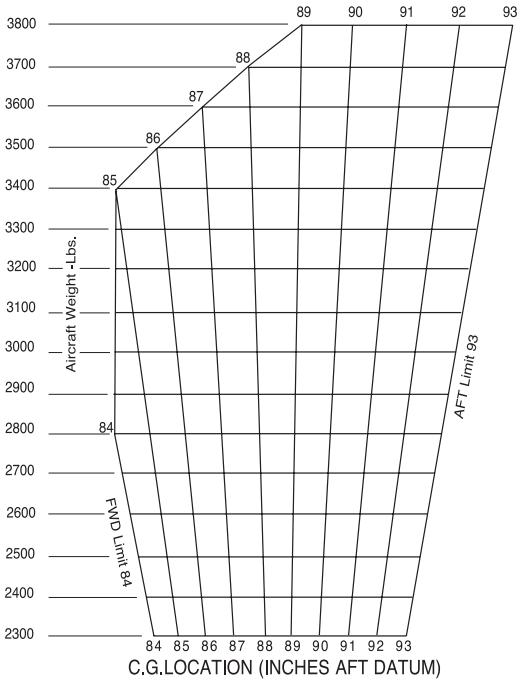
Plot Zero Fuel CG and Takeoff CG on the CG Envelope Graph Below

Answers: (1) 91.6 (2) 68.4 gal. (3) 3,800 lbs. (4) 91.9 (5) Forward

Formulas

- Weight × Arm = Moment
- Total Moment ÷ Total Weight = CG
- Max Ramp Weight – Zero Fuel Weight = Usable Fuel Weight
- Fuel Weight ÷ 6 = Fuel Gallons
- 100 LL (Blue) Fuel Weighs 6 lbs./gal.; Oil Weighs 7.5 lbs./gal.
- 2 Gallons of unusable fuel and oil at full capacity are Included in Basic Empty Weight

CG Envelope Graph



DEPARTURE PROCEDURES

Passenger Briefing

1. Safety Belt/Harness Usage
2. Cockpit Door Operation
3. Emergency Exit Operation
4. Fire Extinguisher Location/Usage
5. No Smoking
6. PIC Authority/Training/Checkride

Pre-Takeoff Briefing (Standard Procedures)

Engine failure or abnormality prior to rotation:

- Abort takeoff - Throttles immediately closed
- Brake as required - Stop straight ahead

If not enough runway to stop:

- Mixture to cutoff
- Fuel selectors, Magnetos, & Battery Master Off
- Maintain directional control, avoid obstacles

Engine failure after rotation with gear down & sufficient runway remains for a complete stop:

- Maintain directional control
- Throttles immediately closed
- Land straight ahead, brake as required

Engine failure after rotation with gear up and decision made to continue:

- Maintain directional control/Pitch Attitude/Airspeed
- Mixtures, Props, Throttles Full Forward
- Flaps and Gear Up
- Identify with dead foot, Verify by closing the throttle
- Feather Prop
- Mixture to cutoff

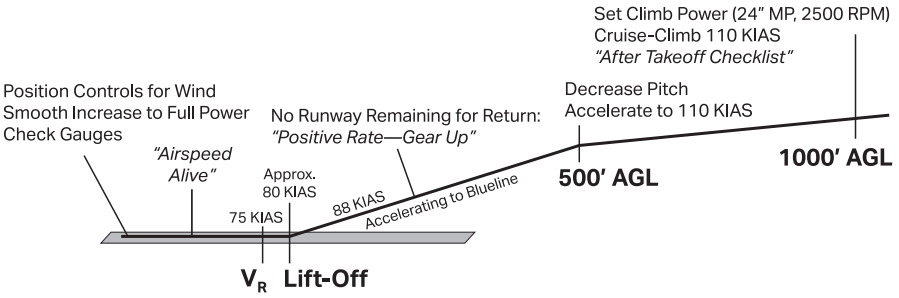
- Climb at 88 KIAS/Blueline
- Declare an Emergency and Land

Normal Takeoff (Flaps 0°)

Do not delay on runway.

1. Approaching centerline, position controls for wind.
2. Smoothly increase throttles to full power.
3. Check engine gauges.
4. *"Airspeed Alive."*
5. Start slow rotation at 75 KIAS.
   (Main gear should lift off at approximately 80 KIAS. 75 KIAS is V_R, not V_{LOF}.)
6. Accelerate to 88 KIAS/Blueline (V_Y).
7. Establish a positive rate of climb and retract the gear if no runway remains.
   (When taking off from runways 4,000' or less, no runway usually remains after a positive rate of climb has been established.)
8. Climb at 88 KIAS/Blueline (V_Y) until 500' AGL.
9. Climb at 110 KIAS after passing through 500' AGL.
   (100-110 KIAS provides better visibility and engine cooling. Density altitude, terrain, obstacles, etc. may require a climb speed from 82-100 KIAS.)
10. Reduce throttles to 24" MP, 2500 RPM at 1,000' AGL.
11. *"After Takeoff Checklist"* out of 1,000' AGL.

Normal Takeoff Profile

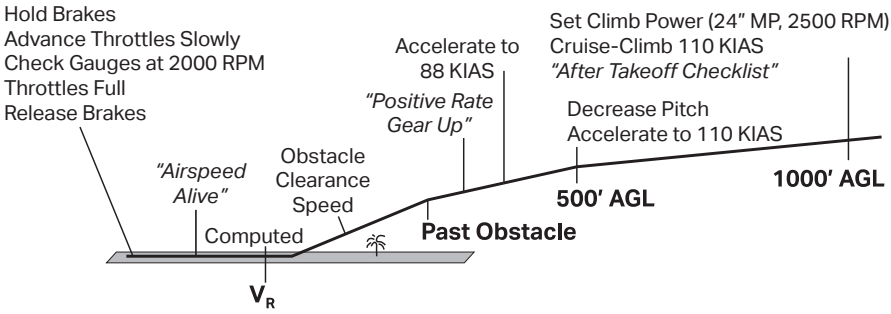


Some ATP aircraft are equipped with Traffic Information System (TIS). TIS is not intended to be used as a collision avoidance system and does not relieve the pilot of responsibility to "see and avoid" other aircraft.

Short Field Takeoff & Climb (Flaps 0°)

1. Hold brakes at departure end of runway.
2. Increase throttles to 2000 RPM.
3. Check engine gauges.
4. Increase throttles to full power.
5. Release brakes.
6. *"Airspeed Alive."*
7. Rotate at 70 KIAS or as specified per Short Field Effort Chart (POH section 5) for Flaps 0° and actual weight.
8. Climb at 82 KIAS through obstacle height or as specified per Short Field Effort Chart (POH section 5) for Flaps 0° and actual weight.
9. Establish a positive rate of climb and retract the gear.
10. Climb at 88 KIAS/Blueline (V_Y) when clear of obstacles until 500' AGL.
11. Climb at 110 KIAS after passing through 500' AGL.
12. Reduce power to 24" MP, 2500 RPM at 1,000' AGL.
13. *"After Takeoff Checklist"* out of 1,000' AGL.

Short Field Takeoff & Climb Profile



Perform all training flight takeoffs with flaps up. ATP prohibits a flaps 25° short field takeoff for training and flight checks due to critically low airspeeds in the event of an actual engine failure. For short field takeoff training, rotate at 70 KIAS, climb at 82 KIAS through 50' or obstacle clearance as minimum speeds for training and flight checks, unless specified by the Examiner to use computed speed for actual weight.

Engine Failure Procedure

After Takeoff with No Runway Remaining and Gear Up

Should an engine failure occur after takeoff and gear-up in a light twin, the following three second, step-by-step technique is suggested. For liability purposes, the following procedure does not cover all situations due to aircraft weight, OAT, density altitude, and aircraft performance. Refer to aircraft POH for additional information.

Engine Failure Occurs – Perform the following in 3 seconds.

One Thousand One: Decrease pitch attitude to horizon or slightly above – approximately 1° on AI. (If the Seminole is trimmed flying out at 88 KIAS and engine failure occurs, the airplane nose will drop towards the horizon due to loss of lift.)

One Thousand Two: Input aileron to bank 2° to 5° into the operating engine. This will assist in directional control and assist in stopping the torque roll effect.

One Thousand Three: Input rudder towards the operating engine. Inputting the rudder after accomplishing the above steps in the first two seconds will help alleviate stomping on the wrong rudder, creating a situation of directional control loss, airspeed loss, and possible altitude loss. Immediately accomplish memory items on "In-Flight Engine Failure" checklist.

EMERGENCY

In-Flight Engine Failure
(After Takeoff)

ATP

For ATP, Inc. training and checkrides only perform Troubleshoot Checklist at or above 3000' AGL. If an actual engine failure occurs, perform Troubleshoot Checklist only if altitude permits.

MAINTAIN DIRECTIONAL CONTROL / PITCH ATTITUDE / AIRSPEED

MIXTURES ..... FULL FWD
PROPS ..... FULL FWD
THROTTLES ..... FULL FWD
FLAPS ..... UP
GEAR ..... UP
IDENTIFY ..... DEAD FOOT
VERIFY/THROTTLE ..... CLOSE
INOP ENG PROP ..... FEATHER
INOP ENG MIXTURE ..... CUTOFF
CLIMB ..... 88 KIAS/BLUELINE

After Takeoff: Climb at 88 KIAS/Blueline or 82 KIAS/V_{XSE} if required for obstacle clearance. Climb to pattern altitude or as required.

DECLARE AN EMERGENCY WITH ATC
LAND AT NEAREST SUITABLE AIRPORT

Perform Engine Failure Secure Checklist if Time Permits

SECTION 5
ARRIVAL PROCEDURES

Approach Briefing

IFR	VFR
Field Elevation	Field Elevation
Type of Approach	Pattern Altitude
NAV Frequency	Type of Landing
Course	Brief 400' AGL altitude for
Glideslope Intercept or FAF Altitude	*"Gear Down—Stabilized"* call
Minimums	
Missed Approach Procedure	

Announced Callouts on Approach

"Gear Down Before Landing Checklist"

Where:

- Visual: Before descending below TPA
- Instrument: ½ dot below glideslope intercept (or at FAF, if no vertical guidance)

Actions - Pilot Flying:

- Complete the Gear Down Before Landing checklist, calling out each step.
- After the first step (GEAR… 3 GREEN), leave your hand on the selector until the PNF also announces *"Three green."* Then, proceed with the remaining steps.

Actions - Pilot Not Flying:

- Monitor the PF's performance of the checklist. (Reference the checklist placard.)
- After the PF announces *"Gear - three green,"* verify that the landing gear is deployed, and then state *"Three green."*

Notes:

- This checklist ***must*** be performed before descending below TPA / the FAF. ***No exceptions.***
- On single-engine training approaches on high-DA days, if descent rates are too high, the PF may reduce drag by deferring flap extension *only*. ***Never defer gear extension*** unless required by an actual emergency.

"Blueline—GUMP. Gas, Undercarriage, Mixtures, Props."

Where:

- Visual: On base or turning final
- Instrument: 1,000 feet AGL

Actions - Pilot Flying:

- Call out *"Blueline—GUMP."*
- Verify the aircraft is on speed (V_{YSE}).
- Check the items, calling out each one in turn (*"Gas, Undercarriage, Mixtures, Props"*).

Actions - Pilot Not Flying:

- Silently back up the PF.

"Gear Down—Stabilized"

Where:

- All approaches: 400 feet AGL

Actions - Pilot Flying:

- Confirm that the landing gear is down and that all stabilized approach criteria are met.
- Announce *"Gear down—stabilized."*

Actions - Pilot Not Flying:

- After the PF's callout, confirm that the gear is down and the approach is stabilized.
- Announce *"Gear down—stabilized."*

Notes:

- If either pilot notes that the gear is not down, or the approach is not stabilized, go around.

"Approaching Minimums" / "Minimums"

- During instrument approaches, to enhance altitude awareness, the PF will also call out *"Approaching Minimums"* when 100 feet above MDA or DA, followed by *"Minimums"* at MDA or DA.

Summary of Callouts

Callout	VFR or Visual Approach	Instrument Approach
PF: *"Gear Down Before Landing Checklist"* PNF: *"Three Green"*	Before descending below TPA (abeam touchdown point, for pattern work)	½ dot below GS intercept (precision) or at FAF (non-precision)
PF: *"Blueline—GUMP. Gas, Undercarriage, Mixtures, Props"*	On base or turning final	1,000' AGL
PF: *"Approaching Minimums"*	N/A	100' above MDA or DA
PF: *"Minimums"*	N/A	At MDA or DA
PF: *"Gear Down—Stabilized"* PNF: *"Gear Down—Stabilized"*	400 feet AGL	

PF = Pilot Flying / PNF = Pilot Not Flying

Stabilized Approaches

The *Airplane Flying Handbook* defines a stabilized approach as "one in which the pilot establishes and maintains a constant-angle glide path towards a predetermined point on the landing runway. It is based on the pilot's judgment of certain visual clues and depends on maintaining a constant final descent airspeed and configuration." Stabilized approaches significantly reduce the chance of landing mishaps.

ATP requires a stabilized approach for all landings, both visual and instrument. Pilots should strive for a stabilized approach throughout their final descent to landing, and the aircraft must be stabilized by the last gear status check (*"Gear Down—Stabilized"*) at 400' AGL. If the approach is not stabilized by that point, or if it becomes unstable later, a go-around is required.

Conditions for a Stabilized Approach

- **Glidepath:** Constant angle glidepath. Proper descent angle and rate of descent to land in the first third of the runway (approximately 450 FPM) must be established and maintained. All available landing aids (ILS, VASI, PAPI, etc.) must be used.
- **Configuration:** Aircraft in landing configuration. Gear down and locked. Flaps and trim in final setting for landing.
- **Airspeed:** Airspeed stable and within ±5 knots of target speed (88 KIAS for normal landings and approaches, 75 KIAS for short-field landings).

These parameters are not merely targets, they are mandatory conditions and limits. Any deviation at or beyond the beginning of the stabilized approach corridor at 400' AGL requires a mandatory go-around.

Go Around Philosophy

The decision to execute a go-around is both prudent and encouraged anytime the outcome of an approach or landing becomes uncertain. ATP considers the use of a go-around under such conditions as an indication of good judgement and cockpit discipline on the part of the pilot.

Go-Arounds During Engine-Out Approaches

In the Piper Seminole, going around on one engine must be avoided. Applying full asymmetric power while at low airspeed poses a serious risk of loss of aircraft control, and the Seminole's limited single-engine climb performance may leave the aircraft unable to climb to a safe altitude for another landing attempt. Per ATP policy, never attempt a single-engine go-around below 500' AGL.

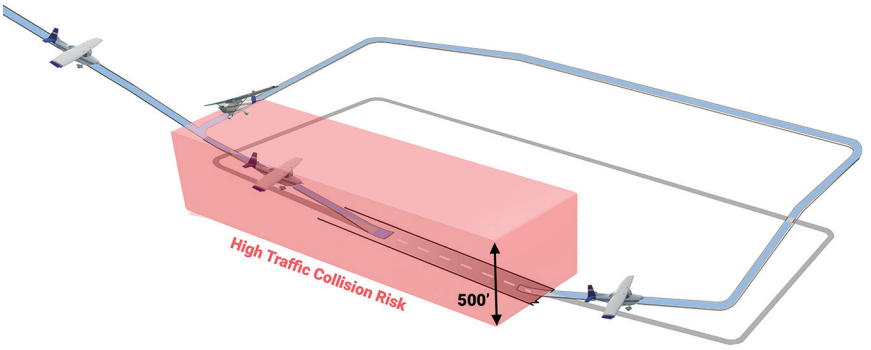
During all simulated engine-out approaches and landings, both engines are available beginning at 400' AGL. Should an unstabilized approach require a go-around, the pilot must smoothly apply full power on both throttles, and take care to maintain aircraft control while the thrust becomes symmetric.

If an actual engine failure has occurred, continuing the approach to a landing is almost always the safer option, even if the approach becomes unstabilized. The pilot should attempt to stabilize the approach as early as possible, and use PIC judgment to choose a course of action most likely to result in a safe landing.

Instructors should vigilantly monitor student approaches and landings, and should command go-arounds if any of the stabilized approach conditions are not met. Instructors should make every effort to avoid allowing a student to take an unstabilized approach close to the ground, requiring the instructor to take the controls and initiate a go-around. Students must also be taught to evaluate their own approaches for the stabilized approach criteria and make the go-around call if necessary, as they alone will be responsible for this during solos, crew cross-country operations, and checkrides.

Collision Avoidance

Pilots need to obtain and maintain situational awareness of other aircraft in, entering, and departing the pattern BEFORE entering the High Traffic Collision Risk box.



Review ATP's **Collision Avoidance Philosophy Supplement** for more information on mid-air collision risk mitigation.

Normal Landing Technique

- Verify that the gear is extended and the approach is stabilized, and announce at 400' AGL: *"Gear Down—Stabilized."*
- Maintain a stabilized descent angle.
- Reduce throttles slowly over approach end, slow to 80 KIAS
- Touch down at approx. 75 KIAS on centerline within the first third of the runway.
- Maintain back pressure on control wheel to prevent slamming the nose wheel onto the runway.

Flap Setting

ATP always recommends landing with flaps 25°. The approach speeds used for flaps 25° are the same as flaps 40° for normal landings. Flaps 25° places the airplane in the best configuration for go-arounds and crosswind landings. Note: Flaps 25° also introduces a secondary gear warning. Use pilot discretion for situations requiring a different setting, such as short field landings, single-engine maneuvering or high winds. The revised PA-44-180 operation manual, section 4.31a states: "landing may be made with any flap setting."

The gear horn is an action horn. The gear warning horn should never be heard below 1,000' AGL on arrivals. Take immediate action by extending the landing gear or by going around.

Normal Visual Approach & Landing

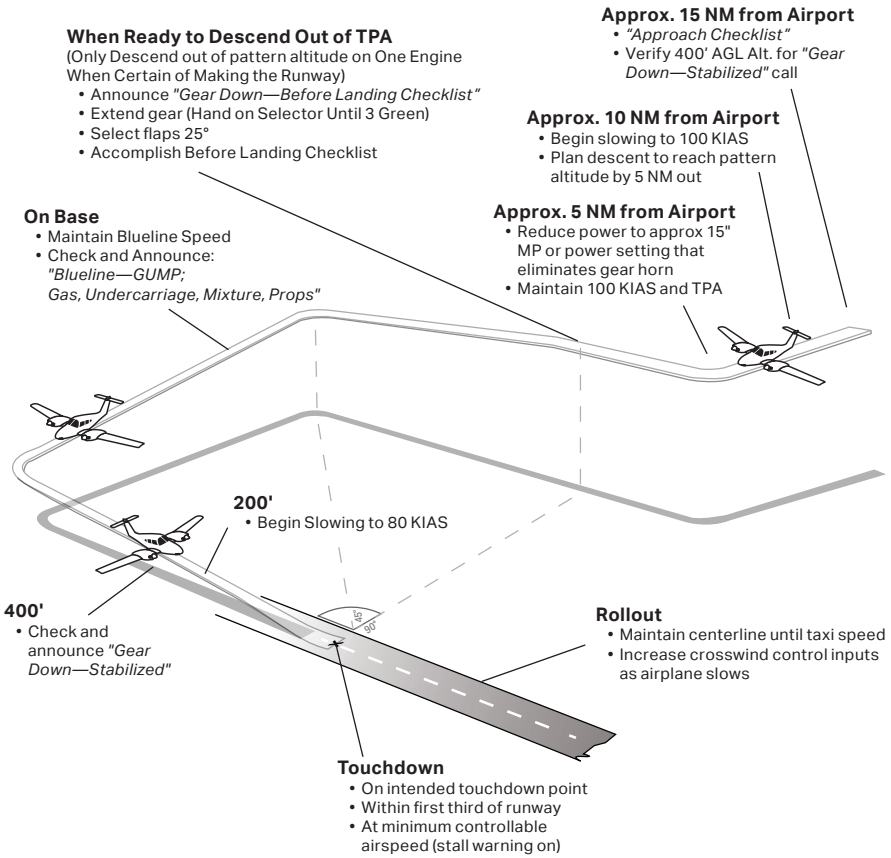
1. Complete the "Approach Checklist" before entering the traffic pattern; devote full attention to aircraft control and traffic avoidance.
2. Slow to 100 KIAS prior to entering downwind or traffic pattern.
3. Enter traffic pattern at published TPA (typically 1,000' AGL).
4. Announce when abeam approach end, on extended base, or on extended final (When ready to descend out of pattern altitude):
   "Gear Down Before Landing Checklist."
5. Extend flaps 25°.
6. Descend out of TPA at 88 KIAS/Blueline.
7. Announce on base leg or turning final: *"Blueline—GUMP."*
   "Gas, Undercarriage, Mixtures, Props."
8. Maintain 88 KIAS/Blueline with flaps 25° on base and final.
9. Announce at 400' AGL: *"Gear Down—Stabilized."*

Whenever leaving TPA you should verify that the *"Gear Down Before Landing Checklist"* has been accomplished. Both pilots must have verified that the gear position indicators show three green.

Gear Down Before Landing Checklist
(ATP, Inc. Max Gear Speed 120 KIAS)

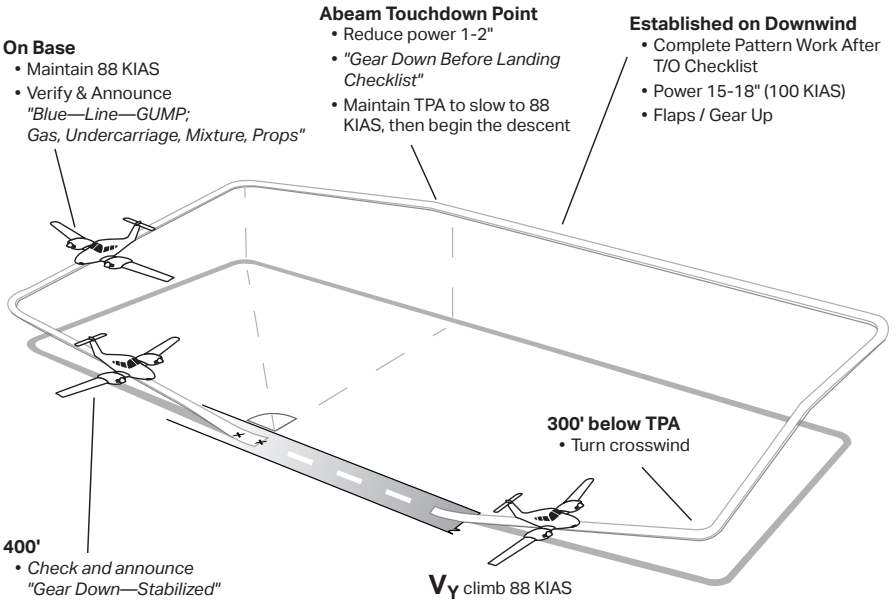
GEAR..... (Hand on Selector Until) 3 GREEN
FUEL SELECTORS.....ON
FLAPS ..... (Vfe 111 KIAS) 25°
MIXTURES..... FWD
PROPS..... FWD
FUEL PUMPS.....ON
Checklist Complete

Normal Visual Approach & Landing Profile



If the pattern altitude is lower than the standard 1,000' AGL, run the Gear Down Before Landing Checklist in its usual position relative to the runway. However, delay the start of descent until you can fly a normal descent path to the intended touchdown point.

Traffic Pattern Operations Profile



Short Field Visual Approach & Landing

1. Complete the "Approach Checklist" before entering the traffic pattern; devote full attention to aircraft control and traffic avoidance.
2. Slow to 100 KIAS prior to entering downwind or traffic pattern.
3. Enter traffic pattern at published TPA.
4. Announce when abeam approach end, on extended base, or on extended final (When ready to descend out of pattern altitude):
   "Gear Down Before Landing Checklist Short Field."
5. Extend flaps 25°.
6. Descend out of TPA at 88 KIAS/Blueline.
7. Announce: (On Base Leg or Turning to Final) *"Blueline—GUMP."*
   "Gas, Undercarriage, Mixtures, Props."
8. On final, extend flaps 40°.
9. Reduce airspeed to speed per Landing Distance Over 50 Ft. Obstacle Short Field Effort Chart (POH section 5) for actual weight. Maintain constant power, airspeed, and angle of descent.
10. Announce at 400' AGL: *"Gear Down—Stabilized."*
11. Close throttles slowly in the flare – touch down with little or no floating.

12. Maintain back pressure on control wheel to prevent slamming the nose wheel onto the runway.
13. Retract flaps after touchdown.
14. Simulate and announce “max braking” for training and checkride purposes.

Use 75 KIAS (max weight speed) on final as a minimum speed for training and flight checks, unless specified by the Examiner to use computed speed for actual weight. To maintain a margin of safety during training, select a touchdown point no earlier than the beginning of the second centerline marking.

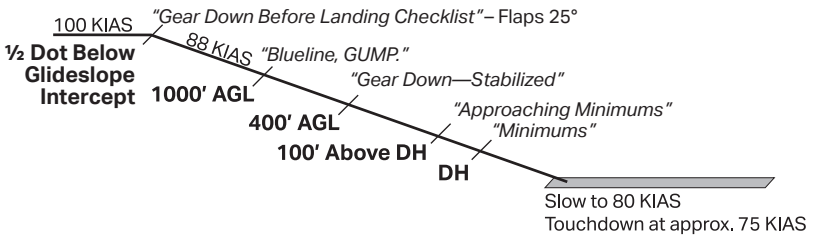
TECHNIQUE: Fly proper speed and maintain some power for round out, close throttles in flare.

Precision Approach One or Two Engines (ILS or RNAV to LPV Minimums)

ATP recommends initially setting flaps 25° at glideslope intercept for both one and two-engine precision approaches. When conducting a one-engine ILS down to minimums it is assumed that a landing is assured. Flaps 25° allows for a stabilized approach to touchdown for both one and two-engine operations. ATP does not recommend a one-engine go-around below 500' AGL. For training purposes, both engines are available from 400' AGL and below, should a go-around become necessary.

1. Complete the *“Approach Checklist”* and identify localizer when able.
2. Set the published inbound course on the HSI.
3. Announce *"Course Alive"* when the localizer begins moving toward center.
4. Check for flags at glideslope intercept altitude and marker.
5. Maintain 100 KIAS until glideslope intercept.
6. Announce *"Glideslope Alive"* when the glideslope begins moving toward center.
7. ½ dot below glideslope intercept: *“Gear Down Before Landing Checklist.”*
8. Extend flaps 25°.
9. Descend on glideslope at 88 KIAS/Blueline (or 100 KIAS considering traffic).
10. Announce out of 1,000' AGL: *“Blueline—GUMP.”*
    “Gas, Undercarriage, Mixtures, Props.”
11. Maintain 88 KIAS/Blueline from 1,000' AGL inbound.
12. Announce at 400' AGL: *“Gear Down—Stabilized.”*
13. Announce at 100' above minimums: *“Approaching Minimums.”*
14. *“Minimums.”*

ILS Approach One or Two Engines Profile



Non-Precision Straight-In Approach One or Two Engines

For a simulated single-engine non-precision straight-in approach: *"Gear Down Before Landing Checklist"* at FAF; maintain 100 KIAS approach speed and flaps 0°; Flaps 25° descending from MDA, followed by the *"Gear Down—Stabilized"* call at 400' AGL. If the airplane will not maintain MDA on a simulated one-engine approach with full power on the operating engine, terminate the approach and use both engines. ATP does not recommend a one-engine go-around below 500' AGL. For training purposes, both engines are available from 400' AGL and below, should a go-around become necessary.

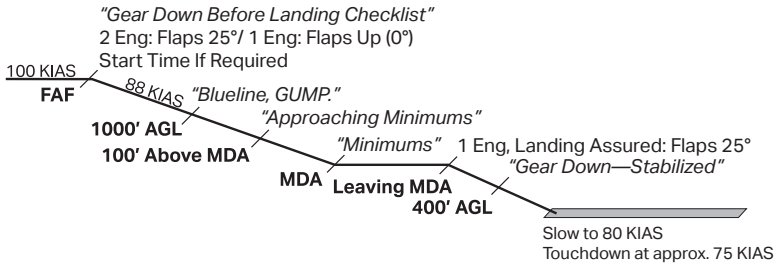
ATP

During simulated single-engine training ATP prohibits gear retraction at MDA unless a missed approach is being demonstrated. With an actual engine failure, you may be required to retract the gear to maintain MDA.

1. Complete the *"Approach Checklist"* and identify the NAV aid when able.
2. Maintain 100 KIAS clean (gear up, flaps up) during procedure turn outbound and inbound to the FAF.
3. Set the published inbound course on the HSI.
4. Announce *"Course Alive"* when the course needle begins moving toward center.
5. Check for flags. 2 miles prior to FAF, verify approach mode on GPS (APR, LPV, LNAV, LP, etc.)
6. Just prior to the FAF: *"Gear Down Before Landing Checklist."*
7. Flaps 0° (one engine) / Flaps 25° (two engines).
8. At the FAF: Start time, Descend at 600-800 FPM at 88 KIAS/Blueline.
9. Announce out of 1,000' AGL: *"Blueline—GUMP."*
   "Gas, Undercarriage, Mixtures, Props."

10. Maintain 88 KIAS/Blueline from 1,000' AGL inbound.
11. Announce at 100' above minimums: *"Approaching Minimums."*
12. *"Minimums."*
13. Maintain MDA (plus 50' minus 0').
14. Runway in sight: descend at predetermined VDP or maintain MDA to MAP.
15. Do not leave MDA until landing is assured.
16. When descending from MDA: Flaps 25° (single-engine)
17. Announce at 400' AGL: *"Gear Down—Stabilized."*

Non-Precision Straight-In Approach One or Two Engines Profile



Circle to Land Approaches Two Engines

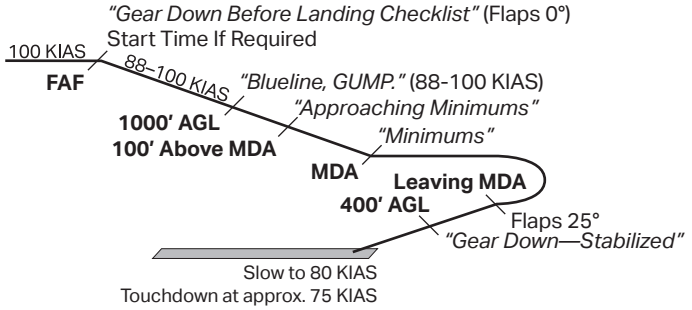
ATP prohibits engine-out (actual or simulated) circle to land approaches in the training environment below Traffic Pattern Altitude.

ATP requires 88 to 100 KIAS approach speed and flaps 0° for circle to land approaches. Unlike straight-in approaches, maintain 88 to 100 KIAS out of 1,000' AGL, descending to MDA. Approach category B is used for minimums. Category B speed ranges from 91 KIAS to 120 KIAS. When leaving MDA, set flaps 25°. Verify and announce *"Gear Down—Stabilized"* at 400' AGL. Do not exceed 111 KIAS (V_{FE}) during approach.

1. Complete the approach checklist and identify the NAV aid when able.
2. Maintain 100 KIAS clean (gear up, flaps up) during procedure turn outbound and inbound to the FAF.
3. Set the published inbound course on the HSI.
4. Announce *"Course Alive"* when the course needle begins moving toward center.
5. Check for flags.
6. Just prior to the FAF: *"Gear Down Before Landing Checklist."*

7. Flaps 0°.
8. At the FAF: Start time, Descend at 600-800 FPM at 88-100 KIAS.
9. Announce out of 1,000' AGL: *"Blueline—GUMP."*
   "Gas, Undercarriage, Mixtures, Props."
   (You may maintain up to 100 KIAS for circle maneuver at MDA.)
10. Announce at 100' above minimums: *"Approaching Minimums."*
11. *"Minimums."*
12. Maintain MDA during circling maneuver (plus 50' minus 0').
13. When turning onto final and descending from MDA: Flaps 25°
14. Announce at 400' AGL: *"Gear Down—Stabilized."*

Circle to Land Approaches Two Engines Profile



Go Around / Missed Approach

Both pilots must watch for unsatisfactory conditions throughout the approach. A go-around procedure must be initiated any time the conditions for a safe approach and landing are not met. Either pilot can command the go-around or missed approach. Some examples of unsatisfactory approach and landing conditions are:

- Unstable approach path or airspeed.
- Improper runway alignment.
- Unexpected hazards on the runway or on final.
- Any sounding of the gear warning horn below 1,000' AGL.
- Anything that jeopardizes a safe approach and landing.

Any time unsafe or unsatisfactory conditions are encountered, a go-around must be immediately executed and another approach and landing should be made under more favorable conditions.

1. Simultaneously increase throttles to full power and pitch for climb.
2. Retract flaps slowly to 0°.
   (Slowly to preclude excessive sink.)
3. Retract landing gear upon acquiring positive rate of climb.
4. Climb at 88 KIAS/Blueline (V_Y) until 500' AGL.
5. Climb at 110 KIAS after passing through 500' AGL (provides better visibility and engine cooling) unless V_X or V_Y is needed for obstacle clearance.
6. Reduce power to 24" MP, 2500 RPM at 1,000' AGL.
7. Announce and perform *"After Takeoff Checklist"* out of 1,000' AGL.

ATP

During simulated engine-out training, both engines are available for a go-around from 400' AGL and below. Do not attempt a single-engine go-around below 500' AGL.

Holding

Max speeds:

Up to 6,000' MSL ..... **200 KIAS**

6,001' to 14,000' MSL ..... **230 KIAS**

1. Slow to 100 KIAS holding speed 3 minutes prior to fix.
2. Acquire EFC.
3. Hold at 100 KIAS, with 1 minute leg to the inbound fix once established.
4. Report altitude and time at holding fix.
5. Make proper entry.

Approach, Procedure Turn & Holding Power Settings

The following suggested power settings are approximations only.

	Speed	2-Eng.	1-Eng.
ILS	88 KIAS (Configured)	15"	20"
Non-Precision	88 KIAS (Configured)	13"	15" (600-800 FPM Descent)
Level at MDA	88 KIAS (Configured)	20"	Full or As Necessary
Holding/PT	100 KIAS (Clean)	18"	Full or As Necessary
Traffic Pattern	100 KIAS (Clean)	18"	Full or As Necessary

SECTION 6

IN-FLIGHT MANEUVERS & PROCEDURES

Spin Awareness

Aerodynamic Factors

All spins are preceded by a stall on at least part of the wing. The primary cause of an inadvertent spin is exceeding the critical AOA while applying excessive or insufficient rudder and, to a lesser extent, aileron. Asymmetrical power could contribute to a spin.

Flight Situations when Unintentional Spins may Occur

- During a V_{MC} Demonstration, when at the point of reaching both V_{MC} and stall simultaneously.
- While maneuvering during slow flight.

Procedure for Recovery from Unintentional Spins

This procedure is taken directly from the Piper Seminole POH.

1. Throttles–Retard to idle.
2. Rudder–Full opposite to direction of spin.
3. Control Wheel–Release back pressure.
4. Control Wheel–Full forward if nose does not drop.
5. Ailerons–Neutral.
6. Rudder–Neutralize when rotation stops.
7. Control Wheel–Smooth back pressure to recover from dive.

ATP Intentional spins are prohibited in multi-engine aircraft. For further reference concerning stall and spin awareness, consult AC 61-67C.

In-Flight Maneuvers

The following maneuvers (except for the V_{MC} Demo unless CLT restricted) are required for all ratings. Additional procedures required for the specific rating sought are subsequently listed in this section for Airline Transport Pilot, Commercial Multi-Engine Initial or Add-On, CFI Multi-Engine Add-On and Private Multi-Engine Add-On.

Clearing Turns for In-Flight Maneuvers

1. Maintain selected altitude.
2. Set bug to entry heading.
3. Lower the wing in the direction of the turn to clear the area below the wing, then roll back to wings level. Remember: "Look Left, Clear Left, Turn Left."
4. Perform 90° clearing turn to left or right.
5. Perform 90° clearing turn back to original heading.

When slowing for maneuvers or approaches, maintain a power setting that eliminates the gear horn. This may increase the target speed to maintain gear horn silence.

Standard Avionics Configurations

ATP requires that GPS and similar avionics equipment be set up in a standard configuration for the various types of flight operation. This will ensure optimal display of traffic, navigation, heading, and course information. Working with a standard avionics configuration in the training environment increases safety, situational awareness, and proficiency.

The standard pages for each set of avionics equipment in each phase of flight are listed in the Standard Avionics Configuration sheet, available in the ATP Library and ForeFlight Documents.

ALL V_{MC} Demonstration

Not required for ATP rating, unless CLT restricted. The V_{MC} Demo is to be accomplished at or above 4,000' AGL, or as specified by Examiner.

In the Piper Seminole, V_{MC} occurs at a lower airspeed than stall speed. The Seminole will lose power as altitude increases because of the reduced density of the air entering the induction system of the engine. Increase the pitch attitude slowly to reduce the airspeed at approximately 1 knot per second while applying rudder pressure to maintain directional control until full rudder is applied. As the speed decreases, additional aileron input will be required to maintain a maximum of 5° bank toward the operating engine. Recover at the first indication of loss of directional control, stall horn, or buffet.

1. Clearing turns.
2. Flaps-up, gear-up.
3. Mixtures–Enrichen, Props–Fwd, Fuel Pumps–On.
4. Slowly close left throttle while maintaining heading and altitude.
5. Slow to 100 KIAS (approx. 10 KIAS above V_{YSE}).
6. Slowly increase right throttle (operating engine) to full power. Use rudder to maintain directional control and bank up to 5° towards the operating engine.
7. Increase pitch attitude slowly, decrease airspeed at approximately 1 knot per second until full rudder is applied to maintain directional control.
8. Recover at first sign of:
   A. Loss of directional control.
   B. First indication of stall (stall horn or buffet).
9. Recover promptly by simultaneously reducing power sufficiently on the operating engine while decreasing the angle of attack as necessary to regain directional control within 20° of entry heading.
10. Continue recovery by increasing power slowly on operating engine while maintaining an AOA that allows for airspeed to increase to a point where directional control can be maintained with a minimum loss of altitude.
11. Accelerate to 82-88 KIAS.
12. Bring throttles slowly together to 20" MP.
13. *"Cruise Checklist."*

Recovery shall not be attempted by increasing the power on the simulated failed engine for demonstration.

(!) During the V_{MC} Demo, the gear horn will sound due to the left throttle being closed. Do not confuse the gear horn, which beeps at 90 cycles per minute, with the steady stall warning horn–typically the first indication of a stall at which to begin recovery.

ALL Maneuvering with One Engine Inoperative

The maneuver shall be performed above 4,000' AGL and in a position where a safe landing at an airport can be accomplished in the event the engine cannot be restarted. Maintain altitude ±100' unless a minimum sink is required due to airspeed at blueline and extremely high density altitude. Maintain heading ±10°.



Below 3,000' AGL, engine failures may be simulated with the throttle only. At 3,000' AGL or higher, engine failures may also be simulated with the mixture. **Never use the fuel selector to simulate an engine failure.**

ALL In-Flight Engine Failure (Shutdown-Secure) Demo

1. Instructor or examiner simulates engine failure with the mixture.
2. Complete the In-Flight Engine Failure Checklist.
3. Complete the Troubleshoot Checklist above 3,000' AGL (if no obvious damage).
4. If no re-start, continue the In-Flight Engine Failure Checklist and complete the Engine Failure Secure Checklist.
5. Maneuver as specified by instructor or examiner.

When maneuvering with one engine inoperative is complete, perform an airstart (separate maneuver) using the Airstart Checklist.



Perform an actual engine shutdown on either engine only above 4,000' AGL. When performing the engine shutdown and secure maneuver as part of a training event (as opposed to an actual emergency), the items on the Engine Failure Secure Checklist are to be ***simulated*** (touched but ***not*** actuated).

EMERGENCY

In-Flight Engine Failure

MAINTAIN DIRECTIONAL CONTROL / PITCH ATTITUDE / AIRSPEED
MIXTURES ..... FULL FWD
PROPS ..... FULL FWD
THROTTLES ..... FULL FWD
FLAPS ..... UP
GEAR ..... UP
IDENTIFY ..... DEAD FOOT
VERIFY/THROTTLE ..... CLOSE

Troubleshoot

THROTTLE ..... 1"
FUEL PUMPS ..... ON
MAGNETOS ..... ON/CHECK
FUEL SELECTORS ..... ON
CARB HEAT ..... ON, CHECK, OFF
FUEL QTY/PRESSURE OR FLOW ..... CHECK
OIL PRESSURE/TEMP ..... CHECK
If engine inop, continue In-Flight Engine Failure Checklist & Engine Failure Secure

INOP ENG PROP ..... FEATHER
INOP ENG MIXTURE ..... CUTOFF

Engine Failure Secure

OPERATING ENG MIXTURE ..... LEAN AS REQ
TRIM ..... AS REQ
INOP ENG MAGNETOS ..... OFF
INOP ENG FUEL PUMP ..... OFF
INOP ENG ALTERNATOR ..... OFF
INOP ENG FUEL SELECTOR ..... OFF
INOP ENG COWL FLAP ..... CLOSED
REC/BLOW ..... OFF
PERFORM CRUISE CHECKLIST

Example Single-Engine Drift Down Scenario

Inoperative Engine at 8,000' MSL, 20° C, and 3,400 lbs. (This is the approximate weight for two people, bags, and half fuel.) The following computations are based on the "Climb Performance—One Engine Operating—Gear Up" chart in the Seminole POH.

Associated Conditions: Flaps 0°, Operating Engine Cowl Flap Open, Inoperative Engine Cowl Flap Closed, Landing Gear Up, Mixture Full Rich, Inoperative Engine Prop Feathered, Full Throttle, 2700 RPM, 88 KIAS/Blueline. (To achieve maximum single-engine performance, the operating engine mixture should be leaned as required.)

Altitude *(Feet MSL)*	**Temp** *(°C) / (°F)*	**Rate of Climb** *(FPM)*	**Distance at 88 KIAS/Blueline** *(Approx. 1.5 NM per minute)*
8,000	20 / 68	-120	From 8,000 to 7,000: 13 NM
7,000	22 / 72	-80	From 7,000 to 6,000: 19 NM
6,000	24 / 75	-40	From 6,000 to 5,000: 37 NM
5,000	26 / 79	0	Single Engine Absolute Ceiling
4,000	28 / 82	50	Single Engine Service Ceiling

All figures are approximate.

ALL Airstart Procedure

The airstart procedure varies between the 1979, 2000, and 2022 model Seminoles since the 2000 models are equipped with an unfeathering accumulator and the 2022 models are fuel-injected.

1979 Model Seminole

Airstart

INOP ENG FUEL SELECTOR ..... ON
INOP ENG CARB HEAT ..... OFF
INOP ENG MIXTURE ..... FULL FWD
INOP ENG PROP ..... FULL FWD
INOP ENG FUEL PUMP ..... ON
INOP ENG ALTERNATOR ..... ON
INOP ENG MAGNETOS ..... ON
AVIONICS MASTER ..... (if conditions permit) OFF
INOP ENG THROTTLE ..... TWO FULL STROKES
INOP ENG THROTTLE ..... 1/4" OPEN
INOP ENG STARTER ..... (ATP 8 seconds max) ENGAGE

PROP WINDMILLING

INOP ENG STARTER ..... RELEASE

WHEN ENGINE BEGINS TO RUN

THROTTLE ..... INCREASE SLOWLY TO 15" MP
MAINTAIN 15" MP UNTIL CHT GREEN
AVIONICS MASTER ..... ON
PERFORM CRUISE CHECKLIST

ABNORMAL

2000 Model Seminole

Airstart

INOP ENG THROTTLE ..... CLOSED
INOP ENG FUEL SELECTOR ..... ON
INOP ENG CARB HEAT ..... OFF
INOP ENG MAGNETOS ..... ON
INOP ENG FUEL PUMP ..... ON
INOP ENG ALTERNATOR ..... ON
INOP ENG MIXTURE ..... FULL FWD
AIRSPEED ..... 100 KIAS (min) / 120 KIAS (recommended)
INOP ENG PROP ..... FULL FWD
INOP ENG THROTTLE ..... 1/4" OPEN

The prop may not easily come out of feather unless airspeed is at least 100 KIAS. It may take up to 20 seconds for prop to windmill. If prop does not windmill verify that there is no prop movement, then engage starter for up to 8 seconds max.

WHEN ENGINE BEGINS TO RUN

THROTTLE ..... INCREASE SLOWLY TO 15" MP
MAINTAIN 15" MP UNTIL CHT GREEN
PERFORM CRUISE CHECKLIST

ABNORMAL

ABNORMAL

Airstart

INOP ENG THROTTLE ..... CLOSED
INOP ENG FUEL SELECTOR ..... ON
INOP ENG ALT-AIR ..... CLOSE
INOP ENG MAGNETOS ..... ON
INOP ENG FUEL PUMP ..... ON
AIRSPEED ..... 100 KIAS (min) / 120 KIAS (recommended)
INOP ENG THROTTLE ..... 1/4" OPEN
INOP ENG PROP ..... FULL FWD
INOP ENG MIXTURE ..... (after prop rotation) ADVANCE

IF NO PROP MOVEMENT WITHIN 7 SECONDS

INOP ENG MIXTURE ..... FULL RICH
INOP ENG STARTER ..... ENGAGE UNTIL PROP WINDMILLS (max 8 seconds)
Engage starter immediately after advancing mixture to avoid flooding engine.

WHEN ENGINE BEGINS TO RUN

THROTTLE ..... INCREASE SLOWLY TO 15" MP
INOP ENG ALTERNATOR ..... ON
MAINTAIN 15" MP UNTIL CHT GREEN
PERFORM CRUISE CHECKLIST

IF RESTART NOT SUCCESSFUL

INOP ENG MIXTURE ..... CUTOFF
INOP ENG THROTTLE ..... CLOSE
INOP ENG PROP ..... FEATHER
PERFORM ENGINE FAILURE SECURE CHECKLIST

ALL Recovery from Unusual Attitudes

Nose High–Climbing

(Airspeed Decreasing)

1. Increase power.
2. Pitch level.
3. Bank 0°.
4. *"Cruise Checklist."*

Nose Low–Descending

(Airspeed Increasing)

1. Reduce power.
2. Bank 0°.
3. Pitch smoothly level.
4. *"Cruise Checklist."*

PVT Steep Turns

Steep turns are to be accomplished above 3,000' AGL. Roll into a coordinated 360° steep turn with at least 45° of bank.

1. Perform two 90° clearing turns.
2. Set power to maintain 120 KIAS throughout maneuver.
3. Cruise checklist.
4. Set bug to entry heading.
5. Maintain selected altitude ±100'.
6. Maintain 120 KIAS or a safe airspeed not to exceed V_A.

7. Adjust power as necessary during maneuver.
8. Perform 360° turn to the left maintaining at least 45° bank ±5°.
9. Roll out on entry heading ±10°.
10. Perform 360° turn to the right maintaining at least 45° bank ±5°.
11. Roll out on entry heading ±10°.
12. *"Cruise Checklist."*

ACS	Airspeed	Altitude	Bank	Heading
	±10 KIAS	±100'	±5°	±10°

In high density altitude conditions, the reduction in available aircraft performance may require steep turns to be performed at a slower airspeed. Select and announce an appropriate airspeed prior to the maneuver.

PVT Maneuvering During Slow Flight

Slow flight is to be accomplished above 4,000' AGL. As specified by the Examiner, this maneuver may be accomplished in any configuration while demonstrating coordinated straight-and-level flight, climbs, turns, and descents. Establish and maintain an airspeed at which any further increase in angle of attack, increase in load factor, or reduction in power would result in a stall warning (e.g., aircraft buffet, stall horn, etc.).

1. Clearing turns.
2. *"Gear Down Before Landing Checklist."*
3. Extend full flaps 40°.
4. Slow to just above stall warning activation (approximately 60-65 KIAS).
5. Adjust power as necessary to maintain airspeed and altitude while maneuvering.
6. Accomplish straight-and-level flight, climbs, turns, and descents as required, without activating a stall warning.
7. Recover with max power.
8. Retract flaps slowly to 0°.
9. Accelerate to 82 KIAS (V_X), maintain altitude or climb as specified by Examiner.
10. Retract gear, accelerate to 88 KIAS (V_Y).
11. *"Cruise Checklist."*

ACS	Airspeed	Altitude	Bank	Heading
	+10/-0 KIAS	±100'	±10°	±10°

Stall Recovery

When practicing recovery from a fully developed stall in a multi-engine aircraft, immediately reduce the AOA by aggressively moving the elevator control forward to ensure recovery without worrying about a specific loss of altitude. This is not an approach to stall–it is a stall that has broken and an aggressive reduction in AOA is critical. The ACS places no emphasis on maintaining altitude or limiting loss of altitude.

CAUTION: Fully developed stalls may lead to inadvertent, unrecoverable spins.

"It is not recommended that full stalls be practiced unless a qualified flight instructor is present. A power-off or power-on full stall should only be practiced in a structured lesson with clear learning objectives and cautions discussed." (AFH 13-20)

"No multiengine airplane is approved for spins, and their spin recovery characteristics are generally very poor. It is therefore prudent to practice spin avoidance and maintain a high awareness of situations that can result in an inadvertent spin." (AFH 13-18)

A fully developed stall in a multi-engine airplane is a situation that can "result in an inadvertent spin".

"In order to spin any airplane, a stalled condition needs to exist. At the stall, the presence or introduction of a yawing moment can initiate spin entry. In a multiengine airplane, the yawing moment may be generated by rudder input or asymmetrical thrust." (AFH 13-18)

When recovering with max power, advance throttles smoothly. This assists with symmetrical power increases.

PVT Power-Off Stall

Stalls are to be accomplished above 4,000' AGL, with a stabilized descent approximating a 3° final approach glidepath (400 FPM). Private requires prompt recovery after a fully developed stall occurs. As specified by the Examiner, this stall may be accomplished on a selected heading ±10° or in a 15°-bank turn (max bank 20°).

Always reduce throttles slowly to avoid damage to engines.

When recovering with max power, advance throttles smoothly. This assists with symmetrical power increases.

1. Clearing turns.
2. *"Gear Down Before Landing Checklist."*
3. Extend full flaps 40°.
4. Establish a stabilized 400 FPM descent at 75 KIAS.
5. Maintain selected heading, or establish 15°-bank turn as specified by Examiner.
6. Slowly reduce power to idle.
7. Slowly increase pitch attitude to induce a stall.
8. Acknowledge cues of the impending stall and then recover immediately after a fully developed stall occurs. Simultaneously:
   - Reduce AOA – Lower nose. Do not attempt to maintain altitude.
   - Roll wings level.
   - Smoothly apply max power.
9. Accelerate to V_X or V_Y (82 or 88 KIAS) as required. Adjust trim.
10. Retract flaps slowly to 10°.
11. Establish a positive rate of climb, then retract gear.
12. Retract flaps to 0°.
13. *"Cruise Checklist."*

ACS	Bank	Heading
	±10° *Not to exceed 20°*	±10°

PVT Power-On Stall

Stalls are to be accomplished above 4,000' AGL. Private requires recovery after a fully developed stall occurs. As specified by the Examiner, this stall may be accomplished in the takeoff configuration (Gear Down, Flaps 0°) or the departure configuration (Gear Up, Flaps 0°) while on a selected heading ±10° or in a 15°-bank turn (max bank 20°) ±10°.

1. Clearing turns.
2. Set takeoff or departure configuration as specified by the Examiner:
   A. Takeoff configuration (Gear Down, Flaps 0°).
   B. Departure configuration (Gear Up, Flaps 0°).
3. Mixtures–Enrichen, Props–Fwd, Fuel Pumps–On.
4. Slow to 80 KIAS, liftoff speed.
5. Maintain selected heading or establish a 15°-bank turn, as specified by the Examiner.
6. Transition smoothly to approx. 15° pitch-up while increasing power to 18" MP.
7. Acknowledge the cues of the impending stall and then recover immediately after a fully developed stall occurs. Simultaneously:
   - Reduce AOA – Lower nose. Do not attempt to maintain altitude.
   - Smoothly apply max power.
   - Level wings.
8. Accelerate to 82 KIAS (V_X), establish a positive rate of climb.
9. Retract gear (if extended), accelerate to 88 KIAS (V_Y).
10. *"Cruise Checklist."*

	Bank	Heading
ACS	±10° *Not to exceed 20°*	±10°

COM Steep Turns

Steep turns are to be accomplished above 3,000' AGL. Roll into a coordinated 360° steep turn with at least 50° of bank.

1. Perform two 90° clearing turns.
2. Set power to maintain 120 KIAS throughout maneuver.
3. Cruise checklist.
4. Set bug to entry heading.
5. Maintain selected altitude ±100'.
6. Maintain 120 KIAS or a safe airspeed not to exceed V_A.
7. Adjust power as necessary during maneuver.
8. Perform 360° turn to the left maintaining at least 50° bank ±5°.
9. Roll out on entry heading ±10°.
10. Perform 360° turn to the right maintaining at least 50° bank ±5°.
11. Roll out on entry heading ±10°.
12. *"Cruise Checklist."*

NOTE: In high density altitude configurations, the reduction in available aircraft performance may require steep turns to be performed at a slower airspeed. Select and announce an appropriate airspeed prior to the maneuver.

ACS	Airspeed	Altitude	Bank	Heading
	±10 KIAS	±100'	±5°	±10°

COM Maneuvering During Slow Flight

Slow flight is to be accomplished above 4,000' AGL. As specified by the Examiner, this maneuver may be accomplished in any configuration while demonstrating coordinated straight-and-level flight, climbs, turns, and descents. The airspeed selected is that at which any further increase in angle of attack, increase in load factor, or reduction in power would result in a stall warning (e.g. aircraft buffet, stall horn, etc.).

1. Clearing turns.
2. *"Gear Down Before Landing Checklist."*
3. Extend full flaps 40°.
4. Slow to just above stall warning (approximately 60-65 KIAS).
5. Adjust power as necessary to maintain airspeed and altitude while maneuvering.
6. Accomplish straight-and-level flight, climbs, turns, and descents as required, without activating a stall warning.
7. Recover with max power.
8. Retract flaps slowly to 0°.
9. Accelerate to 82 KIAS (V_X), maintain altitude or climb as specified by Examiner.
10. Retract gear, accelerate to 88 KIAS (V_Y).
11. *"Cruise Checklist."*

ACS	Airspeed	Altitude	Bank	Heading
	+5/-0 KIAS	±50'	±5°	±10°

COM Power-Off Stall

Stalls are to be accomplished above 4,000' AGL, with a stabilized descent approximating a 3° final approach glidepath (400 FPM). Commercial requires prompt recovery at the first indication of an impending stall (e.g. aircraft buffet, stall horn, etc.). As specified by the Examiner, this stall may be accomplished on a selected heading ±10° or in a 15°-bank turn (max bank 20°).

ATP

Always reduce throttles slowly to avoid damage to engines.

When recovering with max power, advance throttles smoothly. This assists with symmetrical power increases.

1. Clearing turns.
2. *"Gear Down Before Landing Checklist."*
3. Extend full flaps 40°.
4. Establish a stabilized 400 FPM descent at 75 KIAS.
5. Maintain selected heading, or establish 15°-bank turn as specified by Examiner.
6. Slowly reduce power to idle while increasing pitch attitude to induce stall (approx. 12°).

7. Acknowledge cues of the impending stall and then recover promptly at the first indication of an impending stall. Simultaneously:
   - Reduce AOA – Lower nose. Do not attempt to maintain altitude.
   - Roll wings level.
   - Smoothly apply max power.
8. Accelerate to V_X or V_Y (82 or 88 KIAS) as required. Adjust trim.
9. Retract flaps slowly to 10°.
10. Establish a positive rate of climb, then retract gear.
11. Retract flaps to 0°.
12. *"Cruise Checklist."*

	Bank	Heading
ACS	±5° *Not to exceed 20°*	±10°

COM Power-On Stall

Stalls are to be accomplished above 4,000' AGL. Commercial requires recovery at the first indication of an impending stall (e.g. aircraft buffet, stall horn, etc.). As specified by the Examiner, this stall may be accomplished in the takeoff configuration (Gear Down, Flaps 0°) or the departure configuration (Gear Up, Flaps 0°) while on a selected heading ±10° or in a 15°-bank turn (Max bank 20°) ±10°.

1. Clearing turns.
2. Set takeoff or departure configuration as specified by the Examiner:
   A. Takeoff configuration (Gear Down, Flaps 0°).
   B. Departure configuration (Gear Up, Flaps 0°).
3. Mixtures–Enrichen, Props–Fwd, Fuel Pumps–On.
4. Slow to 80 KIAS, liftoff speed.
5. Maintain selected heading or establish a 15°-bank turn, as specified by the Examiner.
6. Transition smoothly to approx. 15° pitch-up while increasing power to 18" MP.
7. Acknowledge stall cues and recover promptly at the first indication of an impending stall.
8. Simultaneously reduce angle-of-attack, set max power, and level wings.
9. Accelerate to 82 KIAS (V_X), establish a positive rate of climb.
10. Retract gear (if extended), accelerate to 88 KIAS (V_Y).
11. *"Cruise Checklist."*

	Bank	Heading
ACS	±10° *Not to exceed 20°*	±10°

COM Accelerated Stall

Accelerated stalls are accomplished at an altitude that allows the maneuver to be completed no lower than 4,000 feet AGL. Transition smoothly from cruise attitude to a bank angle of 45°, maintaining coordinated turning flight, while increasing elevator back pressure steadily to induce the stall.

1. Clearing turns.
2. Flaps-up, gear-up.
3. Mixtures–Enrichen, Props–Fwd, Fuel Pumps–On.
4. Slow to 100 KIAS.
5. Slowly reduce power to idle.
6. Establish a coordinated 45° bank turn.
7. Maintain altitude to induce stall.
8. Recover at the first indication of an impending stall.
9. Reduce angle-of-attack, level wings, and set max power.
10. *"Cruise Checklist."*

COM Emergency Descent

1. Perform two 90° clearing turns.
2. Throttles–Slowly reduce to idle.
3. Props–Full Forward.
4. Mixtures – Adjust.
5. Gear–Down (V_{LO} Down = 140 KIAS).
6. Cowl flaps–As Required.
7. Initiate turning descent (bank angle 30°-45°) while scanning for traffic.
8. Maintain 120 KIAS (in training), unless requested by examiner or required by emergency to accelerate to V_{LE} (140 KIAS).
9. Notify ATC/Traffic as appropriate.

ATP When recovering from the Emergency Descent, verify airspeed at or below 109 KIAS (maximum gear retraction speed) before retracting the gear.

Required maneuvers for the CFI Multi-Engine Add-On are the same as for the Commercial Add-On except for the addition of the Airspeed Drag Demonstration.

CFI Airspeed Drag Demonstration

Note power required for each drag item and VSI changes throughout each step.

1. Clearing turns.
2. Mixtures–Enrichen, Props–Fwd, Fuel Pumps–On.
3. Slow to 88 KIAS/Blueline.
4. Extend gear.
5. Add power to maintain 88 KIAS and altitude.
6. Extend flaps 25°.
7. Add power to maintain 88 KIAS and altitude.
8. Extend flaps 40°.
9. Add power to maintain 88 KIAS and altitude.

Above steps are for knowledge of drag to required-power relationships

---

ACS-Required Tasks Below

10. Close left throttle. (Windmilling prop)
11. Add max power on right engine, maintain 88 KIAS and heading. (Note VSI.)
12. Left Cowl Flap–As Required.
13. Flaps up, maintain 88 KIAS. (Note VSI.)
14. Gear up, maintain 88 KIAS. (Note VSI.)
15. Left throttle to 12" MP simulated feather power, maintain 88 KIAS. (Note VSI.)
16. Pitch for 82 KIAS. (Note VSI.)
17. Pitch for 100 KIAS. (Note VSI.)
18. Bring throttles slowly together to 20" MP.
19. Left cowl flap: open or as required.
20. *"Cruise Checklist."*

ATP Steep Turns

Steep turns are to be accomplished above 3,000' AGL and may be performed with two 180° turns or one 360° turn. Roll into a coordinated 45°-bank steep turn. Roll into and out of turns at approximately the same rate.

1. Perform two 90° clearing turns.
2. Set power to maintain 120 KIAS throughout maneuver.
3. Set bug to entry heading.
4. Maintain selected altitude ±100'.
5. Maintain 120 KIAS or a safe airspeed not to exceed V_A.
6. Adjust power as necessary.
7. Perform 180° turn to the left maintaining 45° bank ±5°.
8. Roll out at 180° point ±10°.
9. Perform 180° turn to the right maintaining 45° bank ±5°.
10. Roll out on entry heading ±10°.

ACS	Airspeed	Altitude	Bank	Heading
	±10 KIAS	±100'	±5°	±10°

ATP Approach to Stall in Approach Configuration *(Clean)*

Stalls are to be accomplished above 4,000' AGL. ATP rating requires the pilot to announce the first indication of an impending stall (stall horn). As specified by the Examiner, this stall may be accomplished on a selected heading ±5° or in a 15 to 30°-bank turn.

1. Clearing turns.
2. Set power 15" MP.
3. Flaps-up, gear-up.
4. Mixtures–Enrichen, Props–Fwd, Fuel Pumps–On.
5. Pitch to maintain altitude.
6. Maintain selected heading, or establish a 20°-bank turn as specified by Examiner.
7. Announce first indication of impending stall (stall horn).
8. Recover: Simultaneously reduce angle-of-attack, set max power, and level wings.
9. *"Cruise Checklist."*

ATP / Approach to Stall in Landing Configuration

Stalls are to be accomplished above 4,000' AGL. ATP rating requires the pilot to announce the first indication of an impending stall (stall horn). As specified by the Examiner, this stall may be accomplished on a selected heading ±5° or in a 15 to 30°-bank turn.



Always reduce throttles slowly to avoid damage to engines.

When recovering with max power, advance throttles smoothly. This assists with symmetrical power increases.

1. Clearing turns.
2. *"Gear Down Before Landing Checklist."*
3. Set Flaps 40°.
4. Pitch to maintain altitude.
5. Maintain selected heading, or establish a 20°-bank turn as specified by Examiner.
6. Announce first indication of impending stall (stall horn).
7. Recover: Simultaneously reduce angle-of-attack, set max power, and level wings.
8. Retract flaps slowly to 0°.
9. Accelerate to 82 KIAS (V_X).
10. Retract gear, accelerate to 88 KIAS (V_Y).
11. *"Cruise Checklist."*

Lost Comm Procedure (FAR 91.185)

If in VFR conditions, or if VFR conditions are encountered, squawk 7600, remain VFR and land as soon as practicable.

If in IFR conditions, squawk 7600 and fly:

Route *(First that applies.)*	**Altitude** *(Whichever is highest until descent is required for landing.)*
Assigned	**M**inimum IFR Altitude
Vectored	**E**xpected
Expected	**A**ssigned
Filed	

FAR Review

1. To maintain instrument currency, a pilot must have made six approaches and demonstrated proper holding procedures as well as radial and bearing tracking in the last six months.
2. An alternate is not required if the weather at your destination is forecast to be at least a 2,000' ceiling and visibility of at least three miles. The forecast must be from one hour before to one hour after your estimated time of arrival.
3. If an alternate airport is needed, forecasted weather at ETA must be at least 600' ceiling and 2 miles visibility for a precision approach; an alternate airport that offers only a non-precision approach must be at least 800' ceiling and 2 miles visibility.
4. Reserve fuel of 45 minutes is required for IFR flights; 30 minutes for VFR day flights; and 45 minutes for VFR night flights. This reserve is required in addition to the fuel required to fly to your destination and alternate.
5. VOR limits: 4 degrees for VOT, ground checkpoint and dual check. 6 degrees for an airborne check.

6. VOR equipment must be checked every 30 days.
7. Transponders must be checked every 24 calendar months.
8. Pitot static systems must be checked every 24 calendar months.
9. ELT equipment must be checked every 12 months, after half of the battery life, or after 1 hour of cumulative use.
10. An aircraft used for hire must have a 100 hour and an annual inspection.
11. In order to descend below the DH or MDA, all of the following conditions must exist:
    A. The required flight visibility is met.
    B. The aircraft is in a normal position to land.
    C. (1) The runway environment is in sight – descend to land.
    (2) Approach lights in sight – descend to 100' above touch down zone until runway environment is in sight.
    (3) Descend and land if red terminating bars or red side row bars are in sight.
12. Multi-engine aircraft with a V_{SO} of 61 knots or less, or a gross weight under 6,000 pounds do not have to demonstrate positive single-engine climb performance per FAR 23.
13. The minimum equipment list includes a list of equipment that may be inoperative for a particular phase of flight. If not required to have a MEL, comply with the minimum equipment prescribed by the FAR's.
14. Standard Traffic Pattern Altitude (TPA) for multi-engine aircraft is 1,000' AGL.

Answer The Following Sample Oral Questions Prior To Arriving For Training

1. Recite the V speeds.
2. What is the maximum demonstrated crosswind component?
3. Describe the Seminole PA-44-180 engine.
    A. How many cylinders?
    B. Who is the manufacturer?
    C. What is the horsepower rating?
    D. Does it have fuel injectors or a carburetor?
    E. Is the engine turbo-charged or normally aspirated?
    F. Why is the right engine labeled LO-360?
    G. How are the cylinders arranged?
    H. How is ignition provided?
    I. What are the minimum and maximum oil capacities in the 1979 and 2000 model Seminoles?

4. Describe the propeller system.
   A. Who makes the propellers?
   B. What does oil pressure do to the propeller?
   C. Which lever manipulates oil pressure to the propeller?
   D. Which unit regulates oil pressure to the propeller?
   E. What is the function of the nitrogen cylinder?
   F. What is the purpose of the spring in the prop dome?
   G. Define constant speed.
   H. What unit adjusts the propeller to maintain a constant RPM and how does it do it?
   I. Define full feathering.
   J. Will the propeller always feather?
   K. What are centrifugal stop pins?
   L. What is the true purpose of the centrifugal stop pins?
5. What is the correct action for a propeller overspeed?
6. Describe the electrical system.
7. What are the indications of a failed alternator?
8. Will the engines continue to run with the alternator and battery master switches turned off?
9. Describe the vacuum system.
   A. Which instruments are vacuum operated?
   B. What are the normal vacuum operating limits?
   C. How many vacuum pumps does the PA-44 have?
   D. What indications would occur in the event of a vacuum pump failure?
10. Describe the stall warning system.
11. Describe the fuel system.
12. Explain how to cross feed fuel.
13. Describe the landing gear system.
   A. How is the landing gear actuated? Describe the pump.
   B. What keeps the gear in the up position?
   C. What keeps the gear in the down position?
   D. If hydraulic pressure is suddenly lost in flight, what indication, if any, would you have?
   E. In what three situations will the landing gear horn activate?
   F. What unit will not allow the gear to be retracted on the ground?
   G. What is the procedure to extend the gear manually (Emergency Gear Extension)?

H. What airspeed is of importance during manual gear extension?
I. Are the brake and the landing gear hydraulics interconnected?
J. If you lose gear hydraulics, will you still have brakes?
K. What indicates that the gear is in transit and the hydraulic pump is activated?

14. What type of braking system is used by the Seminole? Where is the brake fluid serviced?
15. What type of flaps does the Seminole have?
    A. What are the flap settings on the Seminole?
16. What are the maximum taxi, takeoff, and landing weights?
17. What is the maximum baggage capacity?
18. Define V_{SSE}.
19. What are the drag factors on light twins?
20. Who determines V_{MC} for a particular aircraft?
21. Define V_{MC}.
22. Why is an aft CG used in determining V_{MC}?
23. What are the factors in determining V_{MC}?
24. Define critical engine and list the factors used to determine it.
25. What causes an aircraft to sideslip with the loss of an engine, and what action is required to correct this?
26. How much climb performance is lost when an engine fails?
27. What aircraft equipment checks are required under FAR part 91?
28. Define absolute and single-engine service ceiling.
29. What documents are required to be on the aircraft?
30. Explain lost communications procedures.
31. Will the propeller feather below 950 RPM? Why or why not?
32. Explain the pitot static system.
    A. Does the PA-44 have an alternate static source? If so, how is it activated and what actions are necessary to acquire the most accurate reading?
    B. What instruments are pitot static?
    C. Where is the pitot static port located?
33. How do you prevent a heater overheat?
34. What is the fuel capacity? How many gallons are unusable?
35. What grade fuel is to be used in the PA-44?
36. How many fuel pumps are on the aircraft?
37. When are the electric fuel pumps to be used?

38. What are the various positions on the fuel selector control?
39. Explain the procedure for cross feeding fuel when operating the right engine from the left tank.
40. If an engine failure occurred at 5,000' MSL, or a high density altitude, what would you do to get max performance from the operating engine after performing the In-Flight Engine Failure Checklist?
41. If the cylinder head temp and oil temp approach the caution range, what can be done to assist in cooling?
42. Why does manifold pressure decrease approximately 1" every 1,000' during climb?
43. When an engine is inoperative or feathered, what indication will be observed on the manifold pressure gauge?
44. Why is the manifold pressure gauge not necessarily a good indicator in determining an inoperative engine?

Additional Questions Specific to G500-Equipped Aircraft

1. In the event of an electrical failure, what pilot actions are necessary to ensure continued operation of the standby attitude indicator?
2. If the standby power button on the attitude indicator is not pressed within one minute of electrical failure, what will happen?
3. How do you determine a standard rate turn on an aircraft with a G500 system?
4. In the event of an AHRS failure, which indications will no longer be displayed on the PFD? Which indications will still be visible on the PFD?
5. Which approaches can be accomplished with an AHRS failure?
6. What instrument approaches are available with the PFD inoperative? With the MFD inoperative? With a total PFD/MFD failure?
7. How long does the internal battery last when providing power to the standby attitude indicator?
8. With the loss of the ADC, which lost instrument does not have a backup? What should you do to compensate for the lack of information?

Additional Questions Specific to G1000-Equipped Aircraft

1. If electrical power is lost from both alternators, what other power sources will allow the avionics to continue providing flight instrument information?
2. What are the Line Replaceable Units (LRUs) that make up the G1000 system?
3. What feature of the G1000 should the pilot use in the event of a PFD screen failure?

4. What flight instrument indications would be lost in the event of an AHRS failure? An ADC failure?
5. How is the loss of a flight instrument indicated to the pilot?
6. What is the secondary / backup source of flight instrument information?
7. How are rate of turn and turn coordination displayed on the G1000?


ATP
N245AT
MANIFOLD PRESSURE IN HG ABS
RPM X100
VERTICAL SPEED 100 FEET PER MINUTE
EGT
CARB. HEAT
COWL FLAP
PANEL LIGHTS
GEAR
PIPER



ATPFlightSchool.com
P.O. BOX 1784 • Ponte Vedra Beach, FL 32004-1784
(904) 595-7950 • Fax: (904) 273-2164